



From ensembles to knowledge-augmented reasoning: Information fusion for depression detection in the generative AI era

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ARTICLE INFO

Keywords:

Depression detection
Information fusion
Knowledge-augmented AI
Large language models
Social media mining
Clinical dialogue systems
Multimodal affective computing

ABSTRACT

Computational depression detection has reached a saturation point with standard end-to-end deep learning paradigms. To navigate the rapid proliferation of highly complex models, this systematic scoping review analyzes 87 primary post-LLM studies, spanning both “in-the-wild” social media and structured clinical interactions, published primarily between January 2024 and April 2026. Evaluated alongside 7 foundational 2023 baselines (totaling 94 core methodology papers out of 111 cited references), these works mark a distinct methodological transition from isolated feature engineering to multimodal and architectural synthesis. We introduce a novel taxonomy that categorizes these emerging approaches into four information fusion paradigms: (1) Architecture Fusion, which hybridizes heterogeneous neural components to capture complementary spatiotemporal dependencies; (2) Knowledge Fusion, which injects structured external clinical knowledge (e.g., DSM-5 ontologies) into neural pipelines to bridge the semantic gap; (3) Generative Fusion, which integrates in-context learning (ICL), retrieval-augmented generation (RAG), and parameter-efficient fine-tuning (PEFT); and (4) Data-Level Fusion, which addresses multimodal alignment, dynamic fusion timing, and temporal modeling. Our critical analysis reveals that while zero-shot large language models (LLMs) offer broad semantic reasoning capabilities, fine-tuned encoders and hybrid architectures integrated with domain-specific knowledge currently provide a more rigorously validated balance of reliability, explainability, and performance. We conclude by exposing the “illusion of accuracy” prevalent in proxy-labeled benchmarks and outlining strategic directions for developing clinically valid, privacy-preserving detection systems suitable for real-world clinical workflows.

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1. Introduction

The intersection of Artificial Intelligence (AI) and mental healthcare represents one of the most significant methodological transitions in modern computational medicine. Specifically, the detection of depression via social media mining has evolved from a niche experimental task into a sophisticated computational domain, driven by the exponential growth of user-generated content and rapid advancements in Natural Language Processing (NLP). As digital platforms increasingly become repositories for personal expression, they offer a unique window into the psycholinguistic patterns associated with mental distress. However, the rapid evolution of Generative AI and Large Language Models (LLMs) has disrupted traditional modeling paradigms, necessitating a rigorous re-evaluation of how depression detection systems are architected, validated, and deployed.

1.1. Motivation & the mental health crisis

Major Depressive Disorder (MDD) is a leading cause of global disability, affecting hundreds of millions of individuals across all demographics. This pervasive crisis necessitates scalable and non-invasive screening solutions to bridge the widening gap between the global burden of disease and available clinical resources. Despite the availability of effective treatments, the “treatment gap” remains alarmingly wide, particularly in low- and middle-income countries, where over 75% of individuals receive no care due to resource scarcity and pervasive social stigma [1]. Traditional diagnostic pathways, which rely on retrospective self-reporting and clinical interviews (e.g., PHQ-9, BDI-II), are inherently reactive, episodic, and subject to recall bias.

In response, the past decade witnessed a significant shift toward the computational analysis of user-generated content, initially focusing on passive social media streams (e.g., X/Twitter, Reddit) to identify linguistic markers of mental distress. While this approach promises a scalable mechanism for early screening, social media text in isolation is intrinsically noisy, unstructured, and often lacks verified clinical ground truth. Consequently, recent computational methodologies have broadened in scope to incorporate structured, multimodal clinical interactions, such as human-computer virtual interviews (e.g., DAIC-WOZ), which allow algorithms to capture cross-modal dissonance, such as flat vocal prosody or facial blunting, alongside spoken text. Because a single modality or simple classifier is rarely sufficient to model the complex pathology of MDD across these diverse environments, research has rapidly transitioned toward Information Fusion. This involves the synergistic combination of diverse neural architectures, external knowledge sources, and multimodal data streams to achieve more robust, clinically aligned, and explainable detection [2].

1.2. Historical context (The Pre-LLM era: 2017–2023)

The evolution of computational depression detection has been documented in several prior surveys, which largely cover the “Pre-LLM” era. Dalal et al. [3] and Phiri et al. [4] provide extensive taxonomies of feature extraction methods used between 2013 and 2023, highlighting the dominance of statistical features (LIWC, TF-IDF) and early deep learning embeddings (Word2Vec, GloVe). Similarly, Owen et al. [5] trace the twenty-five-year history of the field, noting the inflection point marked by the introduction of Transformer-based models (BERT, RoBERTa) around 2019.

However, the literature from this period (2017–2023) generally converged on a standard “End-to-End” deep learning paradigm in which

input text is fed into a black-box encoder, such as a CNN, LSTM, or BERT, followed by a classification head. While effective, these methods suffer from two critical limitations:

1. The “Black Box” Problem: They lack the interpretability required for clinical trust, offering only a probability score without a causal explanation.
2. The Reasoning Gap: They rely on statistical correlations, such as n-gram co-occurrence, rather than symptom-based reasoning (e.g., identifying anhedonia or insomnia), which is the primary barrier to clinical integration.

1.3. Relation to previous surveys

While the aforementioned surveys provide excellent overviews of feature engineering and static deep learning, they predate the widespread integration of Large Language Models and knowledge-augmented reasoning in this domain. Unlike previous literature, existing reviews do not address the complex trade-offs introduced by Generative AI, such as hallucination risks and API-based privacy concerns.

This survey differentiates itself by focusing exclusively on the “Post-ChatGPT” era (2024–2026). Unlike predecessors that categorize methods by “feature type” (e.g., linguistic vs. behavioral), we introduce a novel taxonomy based on Information Fusion paradigms. We specifically analyze the shift from single-stream architectures to complex systems that fuse Generative AI, Knowledge Graphs, and Multimodal signals, providing a novel conceptual synthesis of these emerging methodologies, mapped explicitly against clinical data provenance standards.

To clearly distinguish this work from concurrent literature, it is vital to note the limitations of other recent surveys. While recent 2026 reviews, such as Kumari et al. [6], provide broad overviews of traditional machine learning feature extraction across multimodal data (e.g., EEG and speech), and Bhatt et al. [7] synthesize the challenges of cross-platform data integration, they fundamentally predate or omit the paradigm shift toward Large Language Models (LLMs) and Knowledge Graphs. Our survey sets itself apart by exclusively taxonomizing the knowledge-augmented and generative fusion architectures that have superseded these traditional methods, introducing entirely different clinical and ethical trade-offs.

1.4. The Paradigm shift (2024–2026)

This survey diverges from previous reviews by analyzing the explosion of high-complexity methods that seek to resolve the limitations of the Pre-LLM era. The period from 2024 to 2026 marks a clear transition, evolving through four distinct paradigms of engineering complexity:

1. From Feature Engineering to Architecture Engineering (Architecture Fusion): Recognizing that single-model baselines have plateaued, researchers began hybridizing heterogeneous neural networks to capture complementary signals. Recent works have moved beyond using BERT or LSTM in isolation by creating sophisticated architectures, such as coupling Convolutional Neural Networks (CNN) for local feature extraction with Bi-LSTMs for temporal dependencies [8,9], or stacking multiple Transformer variants to form robust, low-variance ensembles [10].
2. From Architecture to Knowledge Engineering (Knowledge Fusion): Recognizing that data alone is often insufficient for clinical validity and addressing the scarcity of labeled clinical data, the field is shifting toward neuro-symbolic and knowledge-aware AI. Recent frameworks explicitly perform Knowledge Integration by injecting external clinical knowledge, such as DSM-5 criteria, Medical Ontologies, or Commonsense Knowledge Graphs, into neural pipelines to guide the learning process and ensure clinical alignment [11,12].

3. The Generative–Discriminative Methodological Transition (Generative Fusion): The advent of Large Language Models (LLMs) has introduced a new form of synthesis. Instead of simple classification, systems now employ LLMs (e.g., GPT-4, Llama-2) to perform “In-Context Learning” (ICL) or to generate reasoning chains that feed into traditional discriminative models. This establishes a synergistic feedback loop where the generative component provides reasoning and data augmentation, while the discriminative component ensures classification precision [13,14].
4. From Single-Stream to Multimodal Synthesis (Data-Level Fusion): Addressing the fundamental heterogeneity of social media, the latest evolution involves bridging semantic, linguistic, and temporal gaps. Architectures now dynamically fuse unstructured text with visual cues, acoustic biomarkers, and longitudinal temporal metadata to capture the cross-modal dissonance characteristic of clinical depression [15,16].

1.5. Research questions

This survey aims to systematize these emerging trends by addressing four core research questions (RQs):

- RQ1 (Architecture Fusion): Do complex hybrid architectures and massive ensembles statistically outperform optimized classical baselines and single-stream Transformers, or do they introduce unnecessary computational overhead?
- RQ2 (Knowledge Fusion): How does the incorporation of structured external knowledge, such as depression ontologies or commonsense graphs, affect the interpretability and reliability of detection models compared to purely data-driven approaches?
- RQ3 (Generative Fusion): How can generative AI models be effectively fused with discriminative architectures to balance the trade-off between reasoning capability (hallucination risk) and classification precision?
- RQ4 (Data-Level Fusion): How does the dynamic integration of heterogeneous modalities (e.g., temporal metadata, acoustic signals) and cross-lingual adaptation bridge the semantic and cultural gaps inherent in diverse user-generated content?

2. Methods

To ensure a rigorous and reproducible analysis of the “Post-ChatGPT” era, we conducted a Systematic Scoping Review in accordance with the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines. The selection process is visualized in Fig. 1.

2.1. Search strategy and data sources

We conducted a comprehensive search across six primary repositories to capture both computational and clinical advancements: IEEE Xplore, ACM Digital Library (for core computer science), Scopus (to cover interdisciplinary journals such as JAMA Network Open, Scientific Reports, and significant advancements in Psychology), and Springer-Link. Crucially, to capture interdisciplinary clinical validation and psychiatric integration, we supplemented the search with premier medical and psychological databases: PubMed/Medline and APA PsycINFO.

The systematic search primarily focused on the period between January 1, 2024, and April 30, 2026, to capture the specific rise of Generative AI and the most recent advancements in multimodal fusion. However, to ensure a rigorous comparative analysis, seven pivotal studies from 2023 (e.g., Subramanian et al. [17], Recharla et al. [18]) were explicitly included through backward snowballing. These studies serve as necessary state-of-the-art (SOTA) baselines that

represent the pre-generative era, allowing us to contextually benchmark the 2024–2026 innovations against established architectures. The search strategy utilized a Boolean combination of domain, data, and methodology keywords. The exact search string employed across all databases (adapted with [tiab] tags for PubMed) was:

("Depression Detection" OR "Mental Health Analysis" OR "Suicide Risk") AND ("Social Media" OR "Twitter" OR "Reddit" OR "Weibo" OR "User-Generated Content") AND ("Information Fusion" OR "Multimodal" OR "Ensemble" OR "Neuro-Symbolic" OR "Large Language Model" OR "LLM" OR "Hybrid Architecture")

Selection Criteria and Synthesis: The initial search yielded 1986 records. We applied a multi-stage screening process based on the following inclusion criteria: (1) Studies published between 2024 and early 2026; (2) Methodologies explicitly employing an Information Fusion strategy (Architecture, Knowledge, Generative, or Data-Level); (3) Studies reporting quantitative results on textual or multimodal user-generated data (encompassing both social media platforms and recorded clinical/virtual interviews). Exclusion criteria included: (1) Single-stream baselines (e.g., pure BERT without fusion); (2) Papers focusing purely on clinical biomarkers (e.g., fMRI) without social media text; and (3) Non-peer-reviewed white papers.

After conducting the comprehensive systematic search across both computational and medical databases, removing duplicates, and screening full texts against our strict inclusion criteria, we selected a total of 87 primary studies focusing on information fusion (2024–2026), supported by 7 pre-LLM baseline studies (2023). To ensure full transparency, the 87 primary information fusion studies are explicitly detailed in [Appendix](#) at the conclusion of this manuscript. This comprehensive appendix table catalogs each study's fusion paradigm, dataset validity tier, ethical compliance status, and a quantitative assessment of methodological quality. Note that several additional references, including 2025–2026 systematic scoping reviews and meta-analyses [19–22], are cited extensively throughout the paper to provide clinical benchmarking and contextual background, although they remain distinct from the primary algorithmic fusion analysis. As the objective of this study is to map emerging architectural paradigms rather than to perform a quantitative meta-analysis of effect sizes, the protocol was aligned with scoping review methodologies and was not prospectively registered in PROSPERO.

2.2. Quality assessment and methodological rigor

To evaluate the methodological quality of the 87 included studies and satisfy PRISMA standards for risk-of-bias reporting, we performed a standardized assessment based on two dimensions: (1) Data Validity Tier, where Tier 1 (clinical labels) indicates a lower risk of bias than Tier 3 (proxy labels); and (2) Experimental Rigor, assessing the presence of ablation studies and appropriate baselines.

Data extraction and quality assessment were performed independently by two reviewers. Inter-rater reliability for study inclusion was high (Cohen's $\kappa = 0.88$), with any minor discrepancies resolved through consensus discussion. Informed by the principles of the TRIPOD+AI clinical reporting guidelines [23], each study was categorized using a custom three-point rubric:

- **High Rigor (3/3):** Studies utilizing Tier 1 or 2 clinical or expert-annotated datasets, featuring comparative baselines, and providing sufficient ablation or evaluation protocols.
- **Medium Rigor (2/3):** Studies utilizing self-reported data with standard architectural evaluations but lacking deeper feature ablation or direct clinical validation.
- **Low Rigor (1/3):** Studies utilizing Tier 3 proxy-labeled data with minimal ablation, flagged for a high susceptibility to the “illusion of accuracy”.

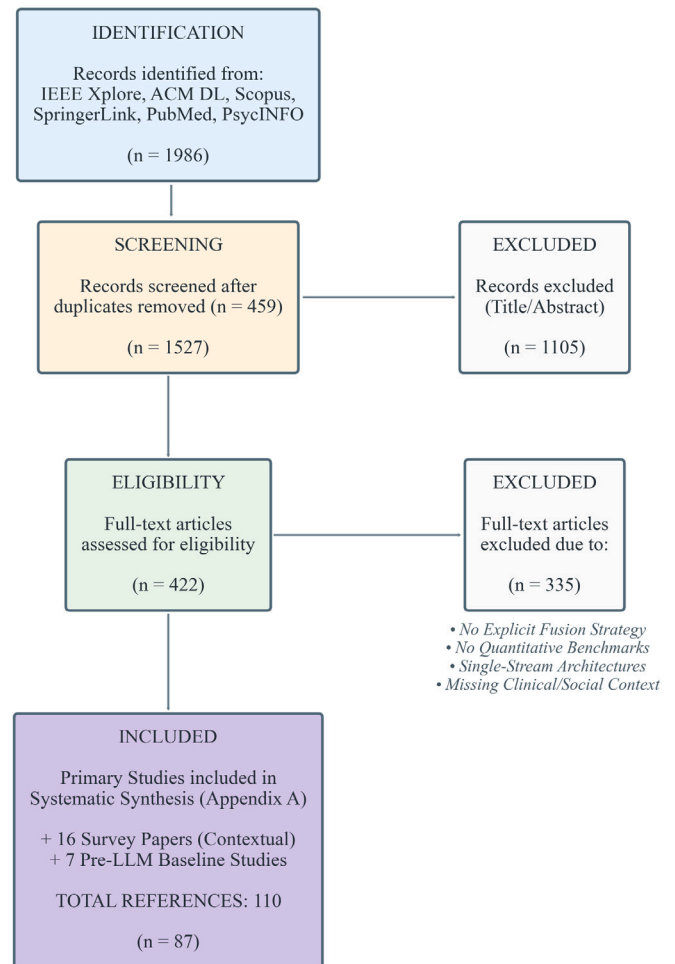


Fig. 1. PRISMA flow diagram illustrating the literature selection process. The updated systematic search identified 1986 records, from which 87 pivotal studies on information fusion strategies were included for final analysis.

The results of this assessment are cataloged in the “Quality / Rigor” column of [Appendix \(Table A.1\)](#). Data extraction utilized a standardized schema to capture five key variables from each study: the primary computational fusion paradigm, the dataset used, the primary evaluation metrics (e.g., F1/Accuracy), the dataset validity tier, and the explicit ethical compliance status regarding data usage. To resolve taxonomic overlap for the quantitative summaries in [Appendix](#), studies utilizing multiple fusion strategies were assigned a single primary paradigm using a formal hierarchical classification protocol (Data-Level → Generative → Knowledge → Architecture), the theoretical justification for which is explicitly detailed in Section 4.

2.3. Survey organization

The remainder of this paper is organized as follows: Section 3 formalizes the depression detection tasks and critically analyzes the landscape of available datasets. Section 4 introduces our proposed taxonomy of Information Fusion strategies, exploring Architecture Fusion (Section 4.1), Knowledge Fusion (Section 4.2), Generative Fusion via LLMs (Section 4.3), and Data-Level Fusion (Section 4.4). Section 5 provides a critical benchmarking of results, explores pathways to clinical integration (Section 5.3), and outlines open challenges. Finally, Section 6 provides concluding remarks.

To visually contextualize the scope of this survey, [Fig. 2](#) illustrates the evolution of detection methodologies. To ensure full transparency, a comprehensive catalog of primary studies categorized by

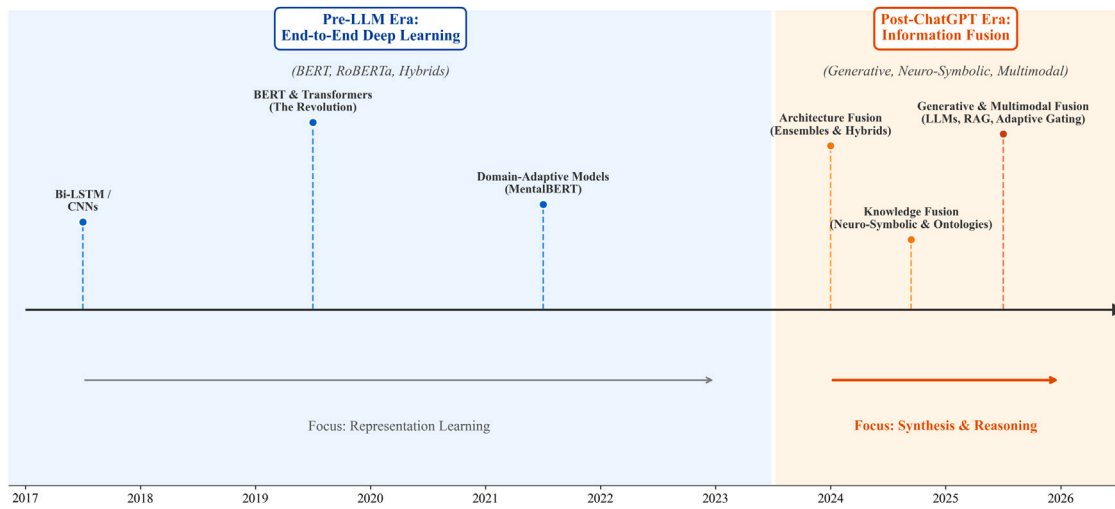


Fig. 2. The evolutionary trajectory of social media depression detection, marking the evolution to Information Fusion (2017–2026).

fusion paradigm, dataset tier, and methodological rigor is provided in [Appendix](#).

3. The evolving landscape of depression detection

To contextualize the emerging fusion strategies, we first formalize the computational tasks and critically analyze the data sources that underpin them. A major finding of this review is that the field suffers from a distinct dichotomy between “High-Volume/Low-Validity” datasets (often used for training data-hungry Transformers) and “Low-Volume/High-Validity” clinical corpora (essential for validating knowledge-augmented systems).

3.1. Problem formalization & task definitions

Depression detection is no longer a monolithic, binary classification task. Recent literature (2024–2026) stratifies the problem into three distinct levels of granularity:

1. Binary Classification (Post-Level vs. User-Level):

- **Post-Level Detection:** This sub-task treats individual texts as independent instances. The objective is to classify a single short text (e.g., a Tweet) as “depressive” or “non-depressive”. However, as noted by Alhamed et al. [24] and Ta et al. [25], this approach often conflates temporary sadness with clinical depression, leading to high false-positive rates.
- **User-Level Detection:** This more rigorous sub-task aggregates a user’s historical timeline to form a longitudinal profile. The goal is to detect depressive subjects rather than depressive content. Recent works, such as Wang et al. [26], demonstrate that user-level modeling captures temporal dynamics (e.g., circadian rhythm disruptions) that are invisible at the post-level.

2. **Severity and Symptom Estimation:** Moving beyond binary labels, this task aligns with clinical triage by categorizing users into severity levels (e.g., Minimal, Mild, Moderate, Severe). Studies by Kerasiotis et al. [27] and Qasim et al. [28] utilize datasets labeled with BDI-II or DSM-5 criteria to perform multi-class classification, arguing that binary detection is insufficient for clinical intervention.
3. **Early Risk Detection (ERDE) & Life Event Mining:** This task introduces a latency penalty. The objective is to detect depression as early as possible [11]. A new sub-variant of this task,

introduced by Lv et al. [29], is Life Event Mining, which seeks to identify specific triggers (e.g., “Divorce,” “Job Loss,”) within the data stream rather than just classification labels, adding causal context to the prediction.

3.2. A critical analysis of the data landscape

The efficacy of any information fusion strategy is fundamentally bounded by the validity of its underlying ground truth. However, as highlighted by recent meta-analyses [22,30], the field suffers from severe heterogeneity in annotation methodologies and clinical validation. We categorize the primary datasets used in the 2024–2026 literature into three tiers based on label provenance, expanding upon dimensions of annotation methodology, sample diversity, and validation rigor (summarized in [Table 1](#)).

- **Tier 1: Clinical and Structured Validation (High Validity):** Datasets in this tier (e.g., DAIC-WOZ, E-DAIC, EATD) are defined by the utilization of validated, standardized psychometric instruments (e.g., PHQ-8, HAMD, SDS) administered or supervised by clinical professionals. The defining characteristic of Tier 1 is its reliance on established clinical rating scales rather than ad-hoc expert opinion. As Fisher et al. (2026) [31] demonstrate, models evaluated on structured clinical interviews exhibit high true-clinical fidelity but generally yield lower raw accuracy scores, reflecting the genuine difficulty of distinguishing overlapping psychiatric symptoms in the real world. Furthermore, clinical validation is often performed by board-certified psychiatrists, ensuring high annotation reliability (e.g., Cohen’s $\kappa > 0.85$).
- **Tier 2: Self-Reported and Expert-Annotated (Moderate Validity):** This tier represents the majority of high-quality social media benchmarks (e.g., eRisk, DepressionEmo, MultiWD). The defining characteristic of Tier 2 is its reliance on subjective self-disclosure or post-hoc human annotation without the use of a standardized clinical rating scale. While experts may review the text, the absence of an instrument like the PHQ-8 prevents strict clinical alignment. This tier includes massive datasets like the one introduced by Shen et al. [32], which is derived from self-disclosure (e.g., “I was diagnosed with...”), as well as specialized corpora like PsyEvent [29] (Life Events) and DepCOV [33] (COVID-19-specific). While these datasets provide massive scale and capture “in-the-wild” linguistic diversity, Tang et al. (2026) [30] warn that they often lack formal psychometric validation, resulting in moderate label noise.

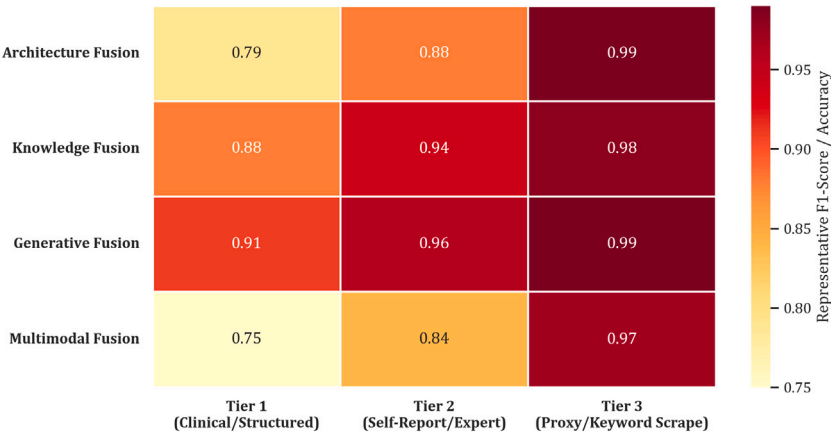


Fig. 3. Performance heatmap illustrating reported effectiveness (F1/Accuracy) across validity tiers and fusion strategies (2024–2026). Scores approaching 1.00 on Tier 3 highlight the “Illusion of Accuracy” inherent in proxy-labeled data, while Tier 1 scores traditionally hover between 0.75–0.91. Note: Generative Fusion via RAG (0.95) represents an emerging clinical outlier. As an aggregate visualization, this serves as a trend indicator rather than a controlled experimental comparison.

- **Tier 3: Proxy-Label Datasets (Low Validity):** This tier encompasses datasets constructed via automated distant supervision, such as scraping posts containing specific keywords (e.g., “sad,” “depressed,”) or using general sentiment polarity (e.g., Sentiment140) as a proxy for clinical depression. This annotation method conflates transient emotional distress with persistent pathological states. We argue that achieving near-perfect accuracy (>99%) on these datasets represents an “Illusion of Accuracy,” as the model merely learns to detect negative sentiment rather than diagnostic criteria.

Crucially, the current data landscape suffers from a profound lack of sample diversity. A recent survey of 108 non-English mental health datasets by Bucur et al. (2026) [34] revealed severe cultural and linguistic biases, noting that clinical expressions of depression vary significantly between individualistic and collectivist cultures. Consequently, fusion models trained exclusively on English-centric Tier 2 or 3 datasets face severe generalizability barriers when deployed globally.

To visually demonstrate the profound impact of data validity on model evaluation, Fig. 3 presents a performance heatmap mapping the four core information fusion paradigms against our dataset tiers. As illustrated, while highly complex models achieve near-perfect scores (~0.99) on Tier 3 proxy data, this performance traditionally drops to more realistic boundaries (0.75–0.91) when tested against Tier 1 clinical ground truths. However, recent breakthroughs in Generative RAG [35] and data-level multimodal fusion [36] are beginning to push these clinical upper bounds toward ~0.95. Nonetheless, this visualization underscores that benchmark supremacy on social media tasks is frequently an artifact of dataset provenance rather than architectural superiority.

4. The proposed taxonomy of information fusion strategies

To systematize the explosive growth of methodologies, we propose a novel taxonomy (see Fig. 4) that harmonizes standard data fusion levels with emerging AI paradigms. Current approaches are categorized into four distinct fusion paradigms:

1. **Architecture Fusion (The Structural Paradigm):** This paradigm focuses on the structural integration of heterogeneous neural components to process spatiotemporal signals.
 - **Feature-Level Fusion:** Concatenating latent vectors from disparate sources. Mahendra and Setiawan [8] demonstrate this by fusing FastText embeddings with CNN-GRU architectures.

- **Decision-Level Fusion (Ensembles):** Aggregates probabilistic outputs. Noor et al. [10] demonstrate that stacking heterogeneous transformers yields statistically significant improvements over single-stream models.
2. **Knowledge Fusion (The Neuro-Symbolic Paradigm):** Representing the shift from correlation to causation, this paradigm injects structured, symbolic knowledge into the neural pipeline.
 - **Ontology Injection:** Integrates external medical graphs to ground neural patterns. Zhang et al. [12] employ a Depression Ontology to map extracted entities to clinical concepts.
 - **Symptom Mapping:** Fuses text with specific diagnostic criteria. Su et al. [39] utilize frameworks that map unstructured text to DSM-5 symptoms.
 3. **Generative Fusion (The LLM Paradigm):** The most recent frontier, where fusion occurs via in-context learning (ICL) or prompt engineering.
 - **Context-Level Fusion:** Uses prompts to fuse user history with LLM reasoning capabilities [14].
 - **Reasoning-Guided Fusion:** Fuses the reasoning power of an LLM with the precision of a discriminative classifier [13].
 4. **Data-Level Fusion (The Multimodal Paradigm):** Addresses the fundamental heterogeneity of the input signals by bridging the semantic gap between text, visual, and temporal modalities prior to or during architectural processing.

While the taxonomy in Fig. 4 and the list above categorize strategies into distinct paradigms for conceptual clarity, contemporary research increasingly blurs these boundaries through cross-paradigm synthesis. For instance, a state-of-the-art system may utilize a large language model (Generative) to extract features, which are subsequently vetted by a knowledge graph (Knowledge), and finally processed by a specialized neural network ensemble (Architecture).

To resolve this taxonomic overlap and avoid arbitrary classifications, this survey employs a dual-classification approach. To illustrate the true architectural complexity of the field, Table 2 provides a multi-label taxonomy matrix, mapping pivotal studies to all overlapping paradigms they utilize. Conversely, to prevent double-counting in our aggregate analyses and Appendix, we apply a strict, formal hierarchical classification protocol. Systems are assigned a single, primary paradigm based on the most upstream technological driver that defines their

Table 1
Comprehensive analysis of primary datasets: Annotation methodologies, clinical validity, and sample diversity.

Dataset	Tier	Annotation & Validation method	Scale (Lang)	Access	Task & Key utilization (2024–2026)
DAIC-WOZ/E-DAIC	Tier 1	Structured clinical interviews via PHQ-8.	~275 users (EN)	Request	Gold-standard for audio-visual-text fusion [15].
EATD-Corpus	Tier 1	Volunteer interviews clinically validated via SDS.	162 users (ZH)	Request	Benchmark for cross-modal attention in Asian cohorts [37].
CLEF eRisk	Tier 2	Expert annotation verified via shared-task criteria.	~1.7k users (EN)	Request	Primary benchmark for knowledge-augmented reasoning [11].
M-HELP	Tier 2	Expert annotated clinical intent.	~2k posts (EN)	Request	Help-Seeking: Distinguishing clinical disorder from general help-seeking.
DepSign-LT-EDI	Tier 2	Domain expert review of user texts.	~7k posts (EN)	Request	Severity: Benchmark for multiclass severity estimation.
PsyEvent	Tier 2	Human-annotated self-disclosure.	~7.9k events (EN)	Public	Life Event: Extracting specific psychological triggers [29].
Shen et al.	Tier 2	Self-reported diagnosis via anchor tweets.	~40k posts (EN)	Public	Post-Level: Training data-hungry ensemble architectures.
WU3D / SWDD	Tier 2	Self-reported patterns and heuristic filtering.	~30k users (ZH)	Request	User-Level: Benchmark for Chinese longitudinal detection.
Nafsany	Tier 2	Expert review of forum posts.	5k posts (AR)	Public	Binary: Low-resource Arabic morphological detection.
RUDA-2025	Tier 2	Expert annotated across Roman/Nastaliq scripts.	15k samples (UR)	Public	Multiclass: Handling code-mixed, low-resource NLP.
DepCOV	Tier 2	Self-reported contextual data.	~10k users (EN)	Public	Pandemic: Isolating COVID-19specific stress triggers.
Suicide & Dep. (Kaggle)	Tier 3	Amalgamated proxy labels from various subreddits.	~232k posts English	Public	Binary: Used to benchmark massive PLM ensembles [38].
Sentiment140	Tier 3	Distant supervision/keyword proxy (No clinical basis).	1.6M posts (EN)	Public	Post-Level: Used for classical ML, but yields ‘Illusion of Accuracy’.

novelty, applied in the following order: Data-Level → Generative → Knowledge → Architecture.

The theoretical justification for this hierarchy is as follows: If a model integrates audio and text, the fundamental input space is altered, making it primarily a *Data-Level* innovation. If the input is standard text but processed by an LLM, the core reasoning engine dictates that it is *Generative*. If it uses standard deep learning but is constrained by external graphs, it is *Knowledge-driven*. Finally, if it relies purely on novel structural combinations of deep networks, it falls to *Architecture*. For example, a system utilizing an LLM (Generative) to process a Knowledge Graph (Knowledge) is coded exclusively as Generative Fusion in [Appendix](#), reflecting the dominant computational engine.

4.1. Architecture fusion: Combining inductive biases

This section investigates RQ1: Do complex hybrid architectures and massive ensembles statistically outperform optimized classical base-lines, or do they introduce unnecessary computational overhead?

We define “Architecture Fusion” as the strategic combination of neural components with distinct inductive biases. As illustrated in [Fig. 5](#), contemporary approaches (2024–2026) typically follow one of four structural sub-paradigms:

1. Sequential Fusion: Feeds static or contextual embeddings into sequence models (e.g., BERT → LSTM).
2. Parallel Feature Fusion: Extracts spatial (CNN) and temporal (RNN) features in parallel streams and fuses them via concatenation or attention.
3. Decision-Level Fusion (Ensembling): Aggregates probabilistic outputs from heterogeneous models to reduce variance.
4. Multi-Task Learning (MTL) Fusion: Utilizes a shared backbone encoder to jointly model depression alongside auxiliary tasks (e.g., demographics) to maximize computational efficiency.

4.1.1. Hybrid deep learning (spatial + temporal fusion)

Before the dominance of large language models, the primary fusion strategy involved coupling CNNs (for extracting local n-gram features) with LSTMs or GRUs (for modeling long-range dependencies).

Recent work has refined this by introducing sophisticated attention mechanisms. Thekkekara et al. [9] proposed the CBA (CNN-BiLSTM-Attention) architecture. By validating on the CLEF eRisk benchmark, they demonstrated that while CNNs capture immediate semantic triggers, the BiLSTM layer is essential for understanding the context of those triggers over time. Their inclusion of an attention mechanism

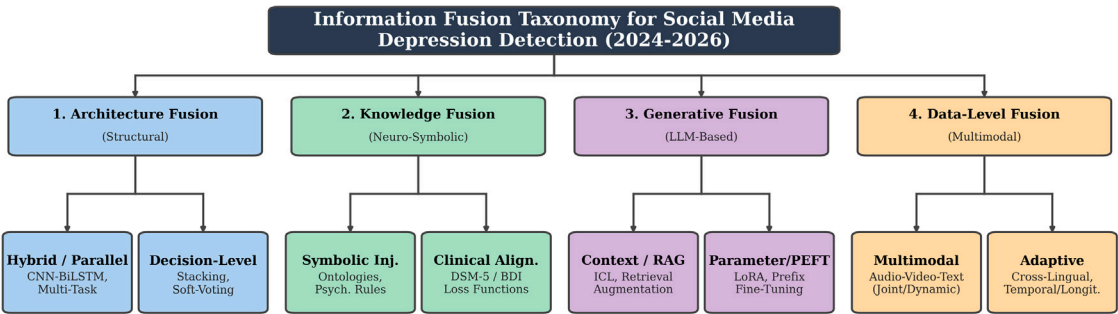


Fig. 4. A hierarchical taxonomy of Information Fusion strategies for social media depression detection (2024–2026). The framework categorizes the current landscape into four core paradigms: (1) Architecture Fusion, encompassing multi-task and parallel structural hybridization; (2) Knowledge Fusion, focusing on neuro-symbolic injection of clinical ontologies and psychiatric rules; (3) Generative Fusion, bridging in-context learning (ICL), RAG, and parameter-efficient fine-tuning (PEFT); and (4) Data-Level Fusion, capturing the technical specifics of multimodal alignment, dynamic fusion timing, and longitudinal temporal modeling.

Table 2
Cross-paradigm taxonomy matrix: Mapping pivotal studies to multiple fusion paradigms.

Ref.	Arch.	Know.	Gen.	Data	Fusion Mechanism & Primary contribution
Noor et al. [10]	✓		✓		Stacking Ensemble: Fuses 6 distinct Transformers via a meta-learner; pure decision-level fusion.
Mahendra and Setiawan [8]	✓		✓		Hybrid Neural: Fuses FastText embeddings with CNN-GRU; purely structural feature integration.
Zhang et al. [12]	✓	✓			Neuro-Symbolic: Fuses neural attention mechanisms with a Depression Ontology.
Dou and Kang [11]	✓	✓			Logic-Injection: Fuses Time-Aware LSTM (Arch) with SenticNet primitives (Know).
Xie et al. [35]		✓	✓		RAG Fusion: Fuses clinical guidelines (Know) with an LLM prompt (Gen), vetted by psychiatrists.
Chen et al. [15]			✓	✓	Multimodal Generative: Fuses Text, Audio, and Video (Data) with GPT-4 (Gen) in a clinical platform.
Shah et al. [40]			✓		PEFT Fine-Tuning: Fuses pre-trained Llama-2 knowledge with domain data via LoRA.
Zheng et al. [14]			✓		ICL Cascade: Uses LLM as a reasoning engine to extract features for a linear classifier.
Hsieh et al. [41]	✓		✓		Prompt Ensemble: Fuses outputs of a single language model guided by multiple persona prompts.
Shen and Paik [42]	✓		✓		Embedding Fusion: Extracts semantic embeddings from Llama 2 (Gen) to feed into a specialized LSTM (Arch).
Li et al. [16]	✓			✓	Dynamic Fusion: Routes Audio and Video (Data) through Mamba/Transformer paths (Arch) based on complexity.
Lan et al. [13]		✓	✓		Gen-Discrim Loop: Uses GPT-4 (Gen) to extract DSM-5 symptoms (Know) for XGBoost classification.
Su et al. [39]	✓	✓			Dynamic Memory: Fuses BERT embeddings with a symptom-update memory bank.

Note: This table utilizes multi-label assignment to illustrate the complex architectural overlap inherent in modern systems. See Appendix A for the single-label, mutually exclusive assignments derived from our Hierarchical Protocol.

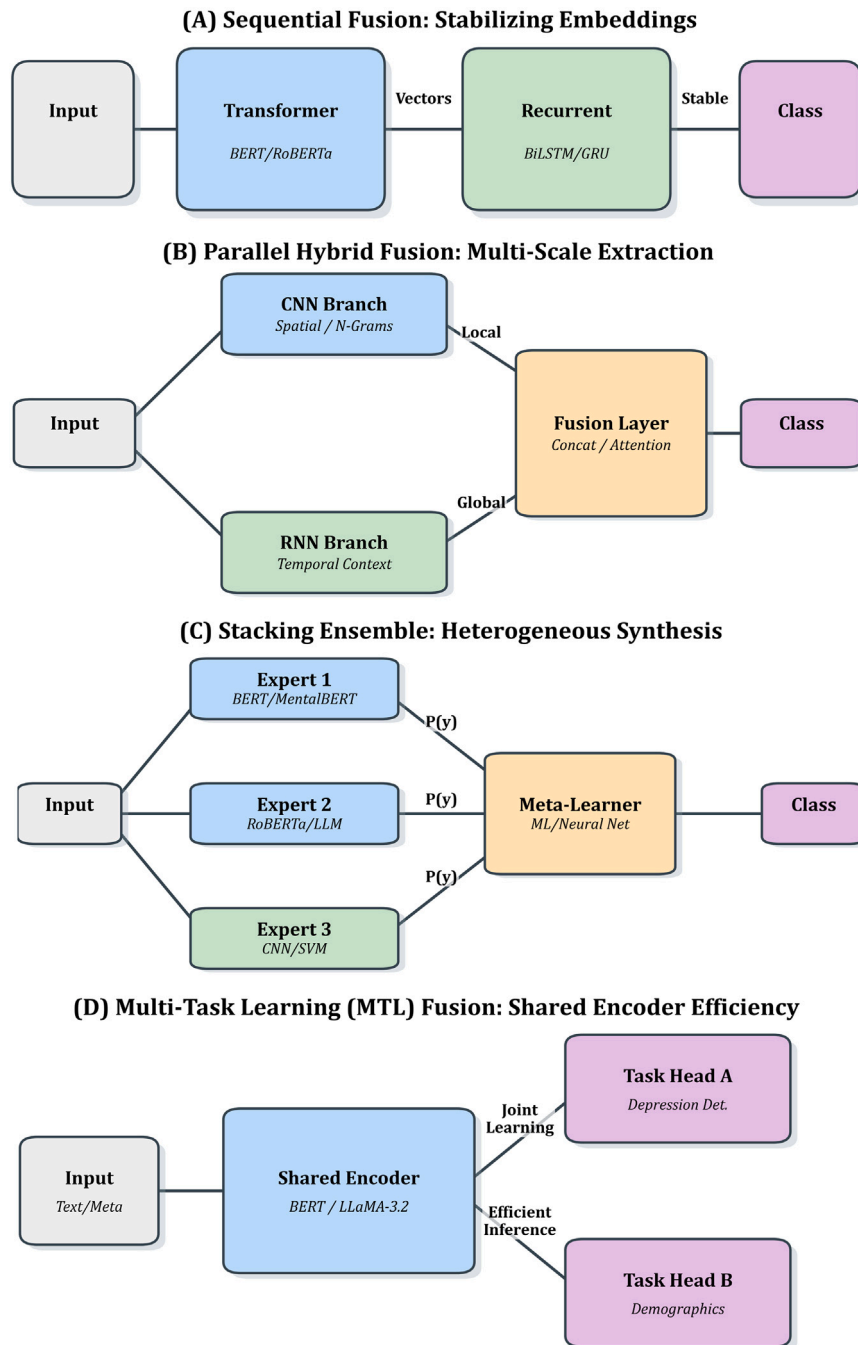


Fig. 5. Taxonomy of architecture fusion paradigms in depression detection: (A) Sequential fusion (Transformer + RNN), (B) Parallel hybrid fusion, (C) Stacking ensemble fusion, and (D) multi-task learning (MTL) with a shared backbone.

specifically improves interpretability, allowing the model to weight personal pronouns (e.g., “I,” “my,”) higher, aligning with psychological theories of self-attentional focus.

Addressing data sparsity, Tejaswini et al. [43] introduced the FCL (FastText + CNN + LSTM) model. By fusing character-level embeddings (FastText) with the hybrid architecture, they mitigated the out-of-vocabulary (OOV) issue inherent in slang-heavy tweets, achieving 88% accuracy. Similarly, Mahendra and Setiawan [8] utilized a CNN-GRU hybrid for Indonesian tweets, highlighting that the CNN layer acts as a feature filter, reducing dimensionality before the sequence modeling phase.

Beyond standard concatenation, the 2026 literature explores more sophisticated mathematical frameworks for feature synthesis. For instance, Dhumane et al. [44] proposed H-Dense FNet, which utilizes

a signal-processing-inspired harmonic analysis layer to fuse a hierarchical attention network (capturing deep semantics) with a DenseNet (capturing handcrafted linguistic features). This shift toward mathematically grounded fusion layers represents an evolution from heuristic integration to formal representational synthesis.

Bridge to Multimodality: Moving beyond text-only fusion, Bin Saeed and Cha [45] introduced DABLNet, which fuses textual embeddings (via BiLSTM) with temporal metadata (posting time) using a cross-modal attention layer. This approach dynamically weights the importance of “what was said” versus “when it was said,” demonstrating substantial reported gains over text-only baselines by capturing behavioral rhythms.

Advancing beyond standard CNN-RNN hybrids, the 2025–2026 literature demonstrates highly specialized structural fusions. Wang et al.

[46] proposed a parallel fusion framework integrating embeddings from two distinct LLMs (BERT and T5) via a logical “AND” operation, maximizing clinical precision. Similarly, Bouktif et al. [47] combined BERT, CNN, and LSTM modules to capture COVID-19-specific suicidal ideation. To optimize computational efficiency, Oulabbib et al. [48] introduced a multi-task learning (MTL) architecture that shares a single backbone (e.g., LLaMA-3.2 or BERT) across depression, age, and gender prediction tasks, reducing inference time by 66%. Addressing morphological complexities, Sheng et al. [49] explicitly fused textual part-of-speech (POS) rules with sequential BiLSTM features for Chinese depression recognition. Furthermore, Villa-Pérez et al. [50] conducted a rigorous comparative analysis, proving that simpler early fusion of psychological lexicons can occasionally outperform complex neural gating mechanisms, depending on temporal data boundaries.

4.1.2. Transformer-RNN hybrids: Sequential embedding fusion

A distinct trend is fusing pre-trained transformers (as dynamic embedding generators) with recurrent neural networks (as classification heads). The hypothesis is that transformers capture global semantics, while RNNs model the sequential decay of emotion.

- **Low-Resource Success:** Baalawi et al. [51] demonstrated that for Arabic text, an ensemble of AraBERT + LSTM/GRU utilized soft-voting to stabilize the high-variance embeddings produced by the transformer, achieving 93.03% accuracy on the Nafsan dataset.
- **High-Resource Redundancy:** Conversely, Louardi et al. [52] performed a rigorous comparison on the English Suicide Watch dataset, revealing that a pure RoBERTa model (97.71%) outperformed hybrid RNN architectures (91.07%).
- **Verdict:** Raj et al. [53] resolved this contradiction, finding that LSTM heads are beneficial primarily for smaller datasets ($N < 10k$) where end-to-end transformers are prone to overfitting.

The 2026 evolution of this hybrid paradigm is exemplified by Sarkar and Majumder [54], who proposed a DeBERTa-BiLSTM architecture for multi-label emotion classification. By fusing the disentangled self-attention of DeBERTa with a history-state fusion BiLSTM, they successfully captured long-range depressive cues that single-stream models often smooth over. Critically, this structural fusion also optimized inference efficiency, reducing time per sample by 26% compared to standard seq2seq architectures, indicating that hybrid heads can mitigate the computational latency of modern transformers.

4.1.3. Ensemble and stacking strategies: Decision-level fusion

To push the upper bounds of performance, researchers employ ensemble learning to address “catastrophic forgetting” and model variance.

Noor et al. [10] presented SENS-Deep, the most complex ensemble in the current literature, stacking six distinct transformers (BERT, RoBERTa, XLNet, etc.) with a RoBERTa meta-learner. While achieving 97.21% accuracy, this approach exemplifies the trade-off between performance and efficiency. A lighter alternative, E-CLS by Tiwari et al. [55], fuses a CNN, an LSTM, and an SVM. This “Shallow-Deep” fusion leverages the SVM’s ability to handle sparse data (TF-IDF) alongside deep semantic feature extraction. Further exploring this high-dimensional space, Baydili et al. [38] demonstrated that ensembles of pre-trained language models (PLMs) perform best when fused with rigorous feature selection algorithms (CWINCA), achieving near-perfect accuracy by pruning redundant embedding dimensions.

Recent studies continue to push the boundaries of decision-level fusion. Kasap et al. [56] demonstrated that a stacking-FFNN ensemble combining five domain-specific transformers (e.g., Mental-BERT, Mental-RoBERTa) consistently mitigates class imbalance in multi-emotion classification. Addressing domain shift, Arshad et al. [57] employed a multi-platform soft-voting ensemble (combining GRU-Attention, CNN, and random forest) to achieve robust generalizability across Twitter, Reddit, and Facebook. Most innovatively,

rather than stacking different architectures, Hsieh et al. [41] introduced a “Multi-Perspective Prompt Ensemble,” which fuses the outputs of a single language model guided by multiple persona prompts (e.g., sociologist, psychologist), aided by auxiliary self-supervision.

To synthesize the diverse methodologies discussed in Sections 4.1.1 through 4.1.3, Table 3 provides a comparative analysis of these fusion strategies, highlighting the specific inductive biases (e.g., locality vs. sequentiality) captured by each architecture and offering a critical assessment of their trade-offs.

4.1.4. The “classical vs. Deep” trade-off: Efficiency as a metric

Despite the dominance of deep learning, a subset of 2024–2026 literature argues for classical machine learning (ML) when computational efficiency is prioritized.

Ding et al. [58] provided a critical trade-off analysis, showing that while ALBERT achieves marginally higher F1-scores than logistic regression, it requires exponentially more training time (see Table 4). This answers RQ1: massive ensembles introduce “unnecessary overhead” for simple binary screening. This trade-off is compellingly illustrated by Kallstenius et al. [59], who performed a rigorous multi-model evaluation on over 51,000 social media posts. For this specific multi-class English dataset, a classical NLP model utilizing an SVM with advanced TF-IDF feature engineering achieved 95% accuracy, outperforming both a fine-tuned LLM (91%) and a prompt-engineered zero-shot LLM (65%). While boundary conditions such as language and task type apply, this indicates that massive generative models may not always be strictly necessary for well-defined, static classification tasks.

Supporting this “lightweight” thesis, Nongkynrih et al. [60] successfully deployed an optimized SVM via a mobile Flask API, proving that classical fusion is sufficient for edge devices. Furthermore, Hameed et al. [61] integrated LIME with random forests, arguing that a slightly less accurate but explainable model is clinically superior to a black-box transformer. As summarized in Table 4, the current literature suggests a clear dichotomy: while deep architectures push the theoretical performance ceiling, optimized classical models remain the pragmatic choice for scalable, privacy-preserving deployment.

4.2. Knowledge fusion: The neuro-symbolic shift

This section addresses RQ2: How does the incorporation of structured external knowledge, such as depression ontologies or common-sense graphs, affect the interpretability and reliability of detection models compared to pure data-driven approaches?

While Architecture Fusion (Section 4.1) optimizes the processing of spatiotemporal data signals, it remains bound by the limitations of inductive learning, in which models can only learn patterns explicitly present in the training data. This reliance creates a “Semantic Gap” in which models learn statistical correlations, such as associating general negative sentiment with depression, rather than clinical causality. Knowledge Fusion, representing the neuro-symbolic shift, seeks to bridge this gap by injecting deductive, structured knowledge into the neural pipeline. As illustrated in Fig. 6, these architectures typically operate via a dual-stream mechanism: a neural stream that captures implicit context and a symbolic stream that enforces clinical logic, converging at a central fusion engine to produce both a diagnosis and a justification.

4.2.1. Knowledge-aware & ontology-based systems

The primary challenge in analyzing social media text is interpreting informal user language, including slang and metaphors, through a clinical lens. Pure deep learning models struggle to map terms like “feeling gray” to clinical symptoms without massive labeled datasets.

To address this, Zhang et al. [12] proposed the Deep Knowledge-Aware Depression Detection (DKDD) framework. Unlike standard end-to-end models, DKDD constructs a Depression Ontology using Bayesian

Table 3
Comparative analysis of architecture fusion strategies (2024–2026).

Ref	Architecture type	Biases combined	Dataset	Perf.	Critique
Thekkekara et al. [9]	CBA (CNN-BiLSTM-Attn)	Locality + Sequential	CLEF eRisk	F1: 0.89	Balanced: Attention provides interpretability for long sequences.
Sarkar and Majumder [54]	DeBERTa-BiLSTM	Disentangled Context + Seq.	Reddit	F1: 0.83	Efficiency: Hybrid head reduces inference latency by 26% for multi-label tasks.
Noor et al. [10]	SENS-Deep (Stacking)	Heterogeneous Context	Mental Health	F1: 0.97	Brute Force: SOTA metrics but computationally prohibitive.
Wang et al. [46]	Parallel LLMs (BERT+T5)	Semantic + Syntactic	E-DAIC	F1: 0.90	Precision: Dual-stream logical 'AND' fusion reduces false positives.
Bin Saeed and Cha [45]	DABLNet	Text + Temporal	Reddit	F1: 0.737	Behavioral: Fuses metadata with text for disorder distinction.
Dhumane et al. [44]	H-Dense FHNet	Semantic + Handcrafted	Kaggle	F1: 0.93	Mathematical: Uses Harmonic Analysis to fuse attention and CNN features.
Oulahbib et al. [48]	Multi-Task Shared Encoder	Joint Demographics + Dep.	Reddit	F1: 0.88	Efficient: Reduces inference time by 66% via parameter sharing.
Tiwari et al. [55]	E-CLS (Heterogeneous)	Deep Semantic + Sparse SVM	DepDs	F1: 0.91	Robust: Blends shallow & deep learners for diverse features.
Hsieh et al. [41]	Prompt Ensemble	Multi-Persona Generative	Mul-tiWD	F1: 0.79	Innovative: Single model ensembling personas instead of heavy stacking.
Baalawi et al. [51]	AraBERT-RNN	Monolingual + Sequential	Nafsany	Acc: 93%	Stabilizing: RNN head fixes Transformer variance in low-resource settings.

Table 4
The efficiency-performance trade-off across classical, deep, and generative architectures (Synthesized from Ding et al. [58] and Kallstenius et al. [59]).

Model architecture	Paradigm	Performance	Efficiency/Cost
Logistic Regression (TF-IDF)	Classical ML	F1: 0.936	1× Cost (Base) [58]
SVM (Advanced TF-IDF)	Classical ML	Acc: 95.0%	Highly scalable/Edge deployable [59]
Bi-GRU (Word Embeddings)	Deep Learning	F1: 0.952	≈ 209× Compute Cost
ALBERT (Raw Text)	Deep Learning	F1: 0.965	≈ 2900× Compute Cost
GPT-4o-mini (Fine-Tuned)	Generative PEFT	Acc: 91.0%	High API/Fine-tuning overhead
GPT-4o-mini (Zero-Shot)	Generative ICL	Acc: 65.0%	Low cost, but clinically inadequate

* Note: While deep and generative models push theoretical boundaries, classical architectures with rigorous feature engineering frequently offer superior performance-to-cost ratios for clinical deployment.

networks to map extracted entities (e.g., “insomnia,” “Zolof”) to medical knowledge. By employing a dual-attention mechanism within the fusion layer, the model weights features based on both temporal recency and ontological relevance. Their ablation studies on the eRisk dataset demonstrated that removing the symbolic stream resulted in a significant drop in F1-score (approximately 0.016), validating the core hypothesis of Information Fusion that external knowledge acts as a semantic regularizer, preventing the neural model from overfitting to surface-level noise.

Similarly, Dou and Kang [11] introduced TAM-SenticNet, a framework that fuses the pattern recognition power of DistilBERT with the symbolic reasoning of SenticNet. By mapping text to specific risk factors defined in a knowledge base, the model creates a symbolic representation that runs parallel to the neural embeddings. This fusion allows the model to infer high-level risks (e.g., “suicidal ideation”) from indirect linguistic cues that might be statistically insignificant in a pure neural model.

However, blindly injecting external knowledge can introduce noise. Chen et al. [62] addressed this with the CoKE (Commonsense Knowledge Enhanced) framework. They argued that simply appending knowledge graph embeddings (via COMET) to all posts overwhelms the classifier. Instead, they utilized a “Three-Way Decision” mechanism to categorize posts into “Positive,” “Negative,” or “Uncertain.” Knowledge fusion is selectively applied only to “Uncertain” posts (the boundary region), significantly improving performance on the eRisk benchmarks by focusing computational resources where semantic ambiguity is highest. This represents a sophisticated adaptive fusion strategy where the model dynamically decides when to fuse external knowledge based on confidence thresholds. Expanding beyond traditional ontologies, knowledge fusion can also be derived from targeted sentiment lexicons; for example, Hossain et al. [63] dynamically constructed “opinion embeddings” using external lexical tools, fusing them with BERT to significantly enhance multi-task mental health categorization.

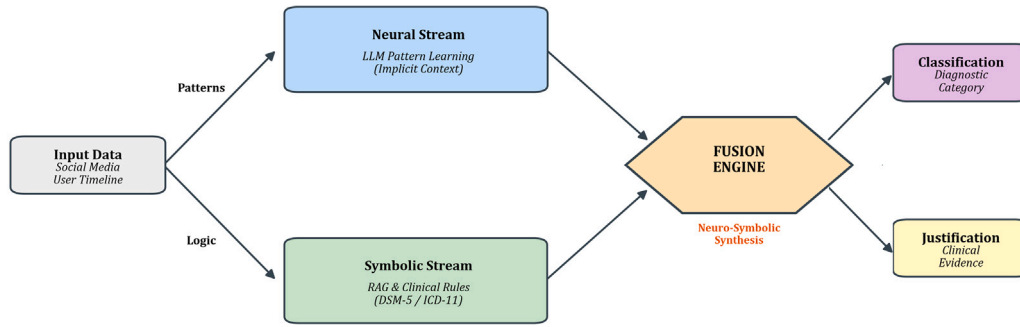


Fig. 6. High-level abstraction of neuro-symbolic fusion. Parallel streams combine neural pattern extraction (LLMs) with symbolic clinical reasoning (RAG, DSM-5). Their integration in a fusion engine produces both diagnostic classification and clinically grounded justification for decision support.

4.2.2. Symptom-aware detection (clinical alignment)

A critical innovation in 2024–2026 is the shift from generic “sentiment analysis” to precise “symptom detection,” fusing unstructured text with the structured schema of diagnostic instruments (DSM-5, BDI-II). A comparative summary of these strategies is presented in Table 5.

- **Structure-Knowledge Fusion:** Several architectures force neural representations to align directly with clinical questionnaires. Recharla et al. [18] introduced W-BERT, which splits the classification head into 21 separate LSTM branches corresponding to the 21 questions of the Beck Depression Inventory (BDI-II). Similarly, Wang et al. [64] proposed E2-LPS, advancing this concept by making the screening process end-to-end differentiable, allowing the model to dynamically learn which posts are clinically relevant (e.g., mentions of hospitalization) rather than relying on static keyword filters. Taking clinical alignment a step further, Zhang et al. [65] designed an “Emotional Pattern Discrepancy Aware” loss function, effectively forcing the neural network to learn the clinical discrepancy between a healthy individual’s emotional response and the affective blunting characteristic of depression, yielding exceptional cross-dataset generalization.
- **Dynamic & Generative Alignment:** Moving beyond static templates, Su et al. [39] proposed a Domain Knowledge Enhancement (DKE) module that fuses BERT embeddings with a dynamic memory bank, updating symptom representations based on incoming data using Multidimensional Item Response Theory (mIRT). Bridging into generative AI, Lan et al. [13] introduced DORIS, a hybrid “Generative–Discriminative” system. Rather than relying on unstable LLM classification, DORIS uses GPT-4 to extract a “mood course” based on DSM-5 criteria, which is then fed into a robust XGBoost classifier. This fusion ensures that the final decision is grounded in statistically robust boundaries while benefiting from the LLM’s semantic extraction capabilities.
- **Granularity & Rare Symptoms:** Validating the need for such granularity, Mendes and Caseli [66] identified that while deep learning excels at abstract signs (e.g., “guilt,”) simple logistic regression often outperforms deep models on concrete somatic symptoms (e.g., “sleep disorder”). Addressing the scarcity of these fine-grained labels, Farruque et al. [67] demonstrated a “Human-in-the-loop” framework that fuses expert annotations with zero-shot learning to harvest rare symptom samples from unlabeled data.

4.2.3. Explainable AI (XAI) integration

In the domain of mental health, a black-box prediction is clinically useless. We categorize post-hoc XAI as a form of interpretability fusion, where the model’s latent vector space is mapped onto a human-readable concept space to fuse the model’s decision boundary with human concepts.

Shah et al. [40] proposed an ensemble model that fuses RoBERTa embeddings with MBTI personality traits (a form of knowledge-feature fusion). Crucially, they integrated LIME (Local Interpretable Model-agnostic Explanations) to visualize which linguistic patterns and personality attributes drove the severity classification. This allows clinicians to audit the model to detect if a “severe” prediction was triggered by genuine risk factors (e.g., “hopelessness”) or artifacts.

Validating the trustworthiness of simpler models, Hameed et al. [61] applied LIME to a GloVe + Random Forest architecture. Their analysis confirmed that even “shallow” models, when fused with high-quality static embeddings, focus on clinically relevant keywords (e.g., “sleep,” “pain”) rather than noise. This reinforces that Knowledge Fusion is not strictly about architecture depth, but about ensuring the model’s focal points align with human clinical intuition.

4.2.4. Clinical practitioner integration and psychiatric validity

As AI architectures achieve near-perfect performance on proxy-labeled social media datasets, recent interdisciplinary reviews highlight a persistent gap in clinical readiness [20,68]. To bridge this, a critical subset of the 2025–2026 literature explicitly incorporates clinical practitioners, such as board-certified psychiatrists, directly into the model development and validation pipelines, moving beyond pure computer science to applied psychiatric informatics.

Rather than relying on generic “negative sentiment,” these studies fuse deep learning with structured psychiatric criteria. For instance, Wan et al. (2026) [69] integrated Cognitive Behavioral Therapy (CBT) principles by pre-training a network to specifically recognize eight types of psychiatric cognitive distortions (e.g., “black and white thinking”). Demonstrating the necessity of clinical nuance, Lam et al. (2026) [70] evaluated large language models on a cohort of patients with alexithymia (emotional blindness). They proved that while self-report scales failed entirely for this subgroup (AUC = 0.35), fine-tuned clinical LLMs maintained robust performance (AUC = 0.95), highlighting the need to account for specific psychiatric comorbidities.

Furthermore, practitioner involvement is increasingly utilized to validate Generative AI frameworks. Xie et al. (2026) [35] developed a retrieval-augmented generation (RAG) system for depression screening that was rigorously annotated and verified by three board-certified psychiatrists using strict DSM-5 criteria, achieving near-perfect clinical consistency ($\kappa = 0.97$). Similarly, Chen et al. (2026) [15] collaborated with psychiatric outpatient clinics to develop a multimodal human–computer interview platform, utilizing GPT-4 alongside the Hamilton Rating Scale for Depression (HAM-D) to achieve diagnostic performance comparable to human clinicians. Beyond direct classification, AI is now being utilized alongside psychologists for scientific discovery. Lissak et al. (2024) [71] utilized a neuro-symbolic pipeline to discover “boredom” as a novel, statistically significant predictor of suicide risk, validating their computational findings against gold-standard clinical scales (CSSRS). Finally, this interdisciplinary transfer of knowledge is actively being deployed in specific medical cohorts, such as leveraging

Table 5
Taxonomy of knowledge fusion strategies (2024–2026).

Ref	Model framework	Symbolic source	Fusion mechanism	Fusion goal
Zhang et al. [12]	DKDD	Depression Ontology	Dual-Attention Layer	Regularize neural features with medical entities.
Dou and Kang [11]	TAM-SenticNet	SenticNet	Symbolic-Neural Concatenation	Infer high-level risks from indirect cues.
Chen et al. [62]	CoKE	COMET (Graph)	Adaptive Gating (3-Way Decision)	Inject knowledge only when model is “Uncertain”.
Hossain et al. [63]	Opinion-BERT	Sentiment Lexicons	Dynamic Embedding Concatenation	Fuse subjective viewpoints with contextual text features.
Recharla et al. [18]	W-BERT	BDI-II	Multi-Head LSTM Initialization	Align outputs with 21 specific clinical symptoms.
Zhang et al. [65]	EPDAD	Clinical Psychology Theory	Discrepancy-Aware Loss Function	Force network to learn affective blunting vs. healthy emotion.
Su et al. [39]	DKE-MUDD	UMID + DKE	Dynamic Memory Bank	Continuously update symptom representations.
Lan et al. [13]	DORIS	DSM-5	Cascade (LLM → XGBoost)	Use LLM for extraction, XGBoost for precision.
Wang et al. [64]	E2-LPS	BDI-II (Templates)	Differentiable Screening	Jointly learn to screen risky posts and detect depression.

transfer learning to adapt social media depression models for breast cancer patients [72], demonstrating the profound impact of fusing AI with clinical oncology and psychiatry (summarized in Table 6).

4.3. Generative fusion: The LLM frontier

This section addresses RQ3: How can generative AI models be effectively fused with discriminative architectures to balance the trade-off between reasoning capability (hallucination risk) and classification precision?

The emergence of large language models (LLMs) in 2024–2026 has introduced a third paradigm of Information Fusion: Generative Fusion. Unlike Architecture Fusion (which merges neural layers) or Knowledge Fusion (which merges graphs), Generative Fusion merges the vast, pre-trained world knowledge of foundation models (e.g., GPT-4, Llama-2) with downstream tasks. This paradigm shifts the engineering bottleneck from “feature extraction” to “prompt engineering” and “instruction tuning.” Our analysis of the literature reveals three distinct strategies for integrating LLMs, as illustrated in Fig. 7: parameter-efficient fine-tuning (PEFT), in-context learning (ICL) Cascades, and Generative Data Augmentation.

4.3.1. Fine-tuning paradigms: Parameter-level fusion

Directly applying general-purpose LLMs to mental health tasks often yields suboptimal results due to domain mismatch. Parameter-Level Fusion addresses this by updating a subset of model weights to fuse the LLM’s general linguistic capabilities with the specific nuances of depressive language.

Shah et al. [40] conducted a pivotal study on this frontier, employing parameter-efficient fine-tuning (PEFT) techniques, specifically low-rank adaptation (LoRA), on Llama2-7B and GPT-3.5 Turbo. Their results demonstrate the potential advantages of this fusion strategy: the fine-tuned GPT-3.5 achieved an F1-score of 0.975, significantly outperforming zero-shot baselines (68.7%) and traditional CNN-BiLSTM models. This highlights that while LLMs possess latent knowledge

of mental health, it must be “unlocked” via fine-tuning to surpass supervised baselines.

Similarly, Shen and Paik [42] utilized Llama 2 not for text generation, but as a superior embedding engine within a temporal modeling framework. By extracting embeddings from Llama 2 and fusing them with a novel “personalized time-difference” feature, they achieved 99.44% accuracy on Twitter data. This reinforces the finding that LLMs serve as powerful semantic encoders, capturing subtle context that older BERT-based variants miss.

However, the cost of this fusion is high. Kuzmin et al. [73] benchmarked multiple generations of models (including Llama 3, Vikhr, and Qwen2) on Russian-language datasets. Their analysis revealed a “performance-resource trade-off”: while fine-tuned LLMs achieved state-of-the-art (SOTA) results on noisy social media text, they barely outperformed lightweight psycholinguistic features on clinically structured essays. This suggests that Generative Fusion is most beneficial when the input data is unstructured and noisy, effectively acting as a semantic denoiser where the LLM’s pre-trained robustness compensates for data irregularity.

A pivotal 2026 study by AlSaad et al. [74] provided a definitive answer to the debate between prompting and adaptation. By benchmarking five prompt-only LLMs against nine fine-tuned encoders on clinical subtyping, they found that task-specific fine-tuning yielded a nearly 20-point gain in Macro-F1 (0.957 vs. 0.765), indicating that for nuanced clinical tasks, parameter-level adaptation generally yields superior metrics compared to general in-context reasoning.

4.3.2. In-context learning (ICL) & reasoning-guided fusion

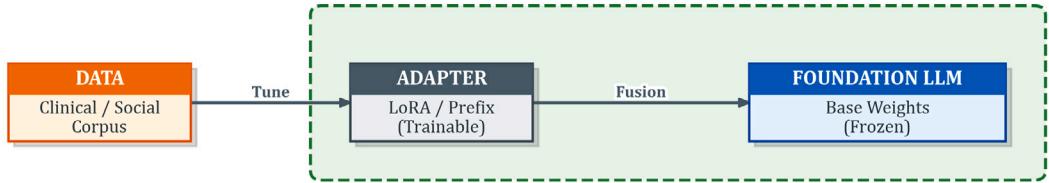
Fine-tuning is computationally expensive and requires accessible weights. Consequently, a parallel stream of research focuses on context-level fusion via in-context learning (ICL), where the fusion occurs in the prompt space.

Zheng et al. [14] introduced the Cascade Large Language Model (CLLM), a seminal example of reasoning-guided fusion. Rather than asking the LLM to classify “Depressed/Not Depressed” (which often leads to hallucinations), they tasked GPT-3.5 with identifying 16 specific behavioral features (e.g., “Insomnia,” “Suicidal Ideation”) via

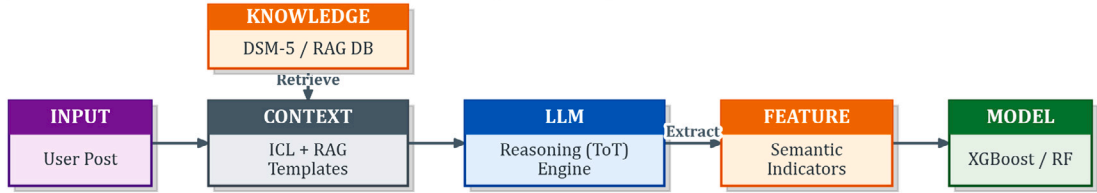
Table 6
Representative 2024–2026 studies featuring direct clinical practitioner involvement and interdisciplinary validation.

Reference	Journal domain	Clinical/Practitioner integration	Psychiatric ground truth
Xie et al. [35]	Psychiatry	RAG outputs rigorously verified and scored by three board-certified psychiatrists.	DSM-5 Criteria
Chen et al. [15]	AI/Medicine	Multimodal interview platform validated by psychiatrists in clinical outpatient settings.	MINI & HAMD-24 Scales
Lissak et al. [71]	Psychiatry	AI pipeline utilized to discover and validate novel suicide risk factors (e.g., boredom).	CSSRS & PHQ-9 Scales
Lam et al. [70]	Medicine	Evaluated LLM robustness specifically on psychiatric subgroups (patients with alexithymia).	HAMD Scale
Wan et al. [69]	Health Informatics	Neural network explicitly pre-trained to recognize Cognitive Behavioral Therapy (CBT) distortions.	SWDD (Clinical Subset)
Fu et al. [72]	Oncology	Transfer learning deployed to screen depression specifically in female breast cancer cohorts.	SDS Scale
Shi et al. [37]	Neuroscience	Multimodal audio-text fusion validated on cohorts assessed via clinical virtual interviews.	SDS Scale

(a) Parameter-Level Fusion (Fine-Tuning / PEFT)



(b) Context-Level Fusion (RAG & Reasoning Cascade)



(c) Data-Level Fusion (Generative Augmentation)

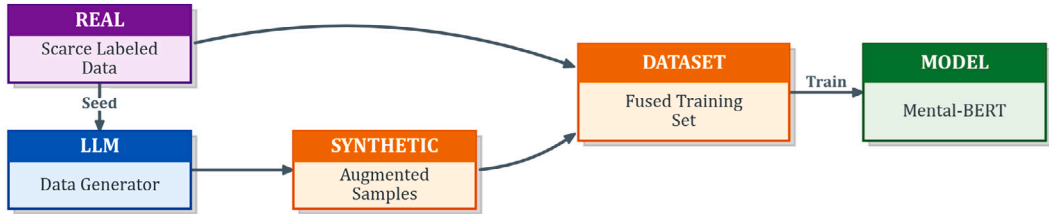


Fig. 7. Classification of Generative Fusion strategies in social media depression detection. (a) Parameter-Level Fusion utilizes Parameter-Efficient Fine-Tuning (PEFT/LoRA) to adapt foundation LLMs to specific clinical domains. (b) Context-Level Fusion employs Retrieval-Augmented Generation (RAG) to ground prompts in external clinical knowledge bases (e.g., DSM-5), using reasoning strategies like Tree-of-Thoughts (ToT) to extract structured features for downstream discriminative classifiers. (c) Data-Level Fusion leverages LLMs to generate high-fidelity synthetic samples, augmenting scarce labeled datasets to improve the robustness of end-to-end models.

Table 7
Comparative analysis of generative fusion (LLM) vs. Discriminative models.

Study	LLM model	Fusion strategy	Baseline	Reported metric comparison	Key finding
Shah et al. [40]	GPT-3.5, Llama2	Fine-Tuning (LoRA)	CNN-BiLSTM	Higher Reported F1/Acc	Fine-tuned GPT-3.5 (F1 0.975) significantly outperformed DL baselines.
AlSaad et al. [74]	Llama-3-8B	Few-Shot (ICL)	Fine-Tuned RoBERTa	Lower Reported F1/Acc	Fine-tuned encoders yielded vastly superior Macro-F1 (+19%) over prompt-only LLMs for subtype classification.
Kallstenius et al. [59]	GPT-4o-mini	Zero-Shot & Fine-Tuned	Classical SVM	Lower Reported F1/Acc	Optimized classical SVM (Acc 95%) beat both fine-tuned (91%) and prompted (65%) LLMs on massive dataset.
Zheng et al. [14]	GPT-3.5 Turbo	Cascade (ICL + Linear)	BERT, BiGRU	Higher Reported F1/Acc	Cascade architecture (Acc 96.17%) beat BERT. CoT reasoning introduced noise.
Shen and Paik [42]	Llama 2	Embeddings (Feature)	RoBERTa	Higher Reported F1/Acc	Llama 2 embeddings captured deeper semantic nuance than RoBERTa.
Kuzmin et al. [73]	Vikhr-7B, Llama 3	Few-Shot & LoRA	Psycholing. Features	Context-Dependent	LLMs won on noisy social media but tied simple features on structured essays.
Wang et al. [75]	Neural-chat-7b	Zero-Shot Prompting	Supervised DL	Higher Reported F1/Acc	Zero-shot LLMs outperformed supervised HAN/LSTM in filling BDI questionnaires.
Sathvik et al. [76]	GPT-4o, Gemini	Zero-Shot	Mental-BERT	Lower Reported F1/Acc	Domain-specific Mental-BERT (F1 60.18%) beat general GPT-4o (F1 41.89%).

ICL. These extracted features were then fed into a lightweight linear neural network for the final classification. This “Cascade” architecture leverages the LLM as a reasoning engine and the linear network as a decision engine, effectively mitigating the over-diagnosis bias observed in pure LLM approaches. Crucially, they found that adding “chain-of-thought” (CoT) prompting actually reduced accuracy (from 96.17% to 95.00%) because the LLM tended to “over-analyze” neutral texts, hallucinating symptoms where none existed. To systematize this reasoning and prevent such unconstrained thought processes, Zhang et al. [77] employed a more structured tree-of-thoughts (ToT) prompting framework within their CALA-Net architecture, forcing the LLM to systematically evaluate text along standardized psychological dimensions before final classification.

Wang et al. [75] applied ICL to the task of clinical questionnaire filling. Their system used zero-shot prompting to map user posts to BDI-II (Beck Depression Inventory) severity levels. By fusing a retrieval mechanism (Universal Sentence Encoder) with the generative model, they enabled the LLM to focus only on symptom-relevant text chunks. Their findings indicate that smaller, instruction-tuned models (e.g., Neural-chat-7b) can outperform larger foundation models when the prompt explicitly constrains the output space to clinical scales.

Addressing longitudinal dynamics, Wang et al. [26] utilized retrieval-augmented generation (RAG) to fuse in-context learning with user timelines. Their framework used RAG to retrieve relevant historical posts to inform the LLM’s summary of a user’s mental health state. While effective for evidence identification, they noted that current LLMs struggle to synthesize long-term temporal dependencies into a coherent timeline summary without architectural modifications, highlighting a limit to current context-level fusion.

4.3.3. Generative data synthesis: Augmentation-level fusion

The third strategy utilizes LLMs to solve the “data scarcity” problem by generating synthetic training samples. This augmentation-level fusion merges real, scarce labeled data with synthetic, abundant data to robustify discriminative models.

Sathvik et al. [76] introduced M-HELP, utilizing LLMs to benchmark help-seeking signals. Their analysis highlighted a critical limitation: general-purpose LLMs (like GPT-4) struggle to distinguish between “demotivation” (a behavioral trait) and “depression” (a clinical disorder) in zero-shot settings, reinforcing the need for domain-specific fine-tuning or augmentation.

To mitigate low-resource constraints, Rajderkar and Bhat [78] employed a translation-based augmentation strategy, fusing English datasets with machine-translated versions in eight Indian languages. While effective for expanding coverage, this approach relies on the assumption that depression manifests linguistically identically across cultures, a limitation noted in cross-cultural studies.

Kerasiotis et al. [27] demonstrated the efficacy of BERT-based augmentation (easy data augmentation). By using a transformer to perform synonym replacement and insertion, they balanced severity classes in the Depression Severity Dataset, improving the weighted F1-score from 72.59% to 84.15%. This confirms that Generative Fusion is particularly potent for addressing class imbalance, one of the field’s most persistent challenges.

A comprehensive comparison of these generative paradigms against traditional discriminative baselines is presented in Table 7. The evidence from 2024–2026 suggests that Generative Fusion is not a universal replacement for discriminative models. While fine-tuned LLMs [40] set new accuracy benchmarks, zero-shot LLMs often fail in precision tasks [76]. The most promising direction appears to be neuro-symbolic cascades [13,14], where the LLM is used to fuse reasoning and feature extraction, but the final decision is delegated to a constrained discriminative layer.

4.4. Data-level fusion & domain adaptation

This section addresses RQ4: How does the dynamic integration of heterogeneous modalities and cross-lingual adaptation bridge the semantic and cultural gaps inherent in diverse user-generated content?

While Architecture and Knowledge Fusion (Sections 4.1 and 4.2) focus on how models process information, Data-Level Fusion addresses

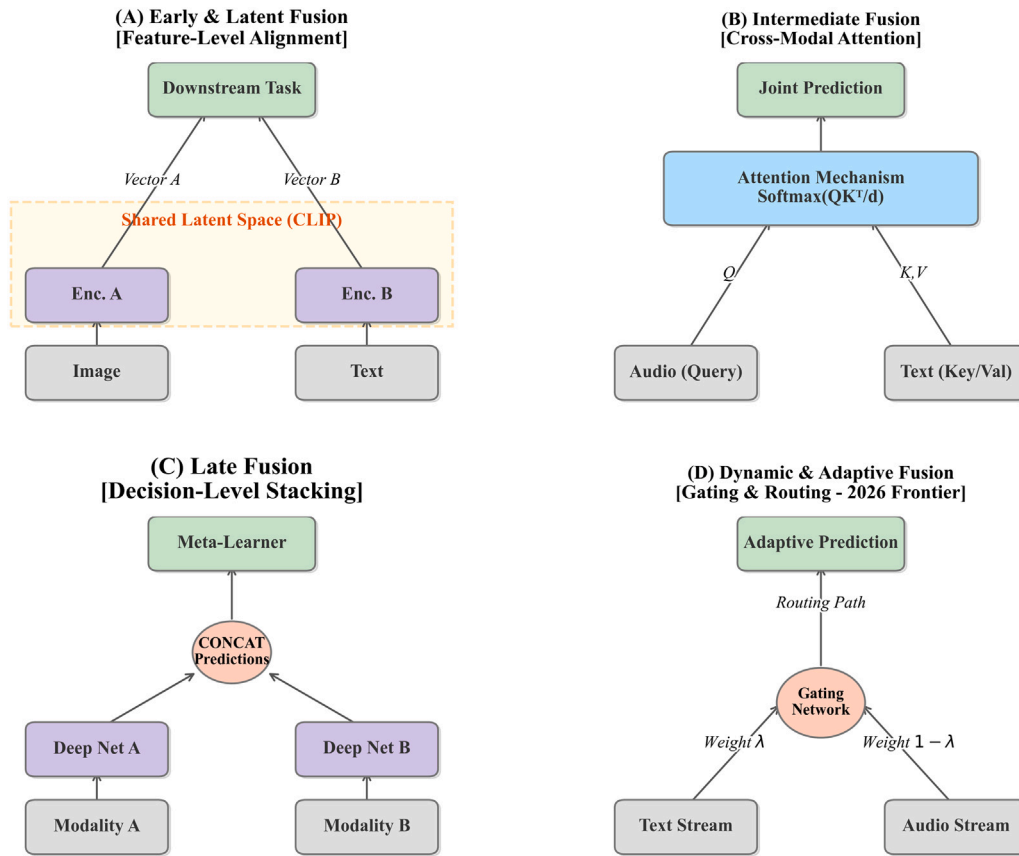


Fig. 8. Visual taxonomy of multimodal fusion timing paradigms. (A) Early and Latent Fusion integrates heterogeneous signals in a shared semantic space (e.g., contrastive CLIP alignment) to capture low-level cross-modal correlations. (B) Intermediate Fusion leverages cross-attention mechanisms to dynamically align and weight independent modality encoders (e.g., Audio-as-Query). (C) Late Fusion aggregates final probabilistic outputs through a meta-learner (stacking), providing superior fault tolerance for scenarios involving missing modalities. (D) Dynamic and Adaptive Fusion represents the 2026 architectural frontier, utilizing gating networks to route data through optimal processing paths based on real-time modality reliability and emotional consistency.

the fundamental heterogeneity of the input signal itself. Social media data is rarely a clean, single text stream; it is multimodal (images and text), culturally diverse (code-mixed languages), and temporally dynamic (time-series). This section analyzes how recent literature (2024–2026) employs fusion strategies to bridge the semantic gap between modalities, the linguistic gap across cultures, and the temporal gap in user histories. This evolution from simple feature concatenation to sophisticated cross-modal reasoning is illustrated in Fig. 8.

Depression is frequently characterized by cross-modal dissonance. For instance, a cheerful image may be paired with a depressive caption, or positive wording may be accompanied by flat vocal prosody. Effectively capturing this dissonance requires precise multimodal fusion. However, as highlighted in the 2025–2026 literature, the effectiveness of these architectures is strongly influenced by fusion timing, namely, the specific network depth at which heterogeneous signals are integrated. We categorize these technical approaches into four fusion timing paradigms, integrating foundational studies from 2024 with the latest breakthroughs reported in 2026.

1. Early and Latent Fusion (Feature-Level): Early fusion involves concatenating raw features or initial embeddings before the main classification network. Xu et al. [79] highlight that, in the context of physiological digital phenotyping, feature-level early fusion risks severe noise propagation if modalities are not strictly synchronized. While computationally efficient, this approach can struggle with unaligned modalities. A more sophisticated variant is latent fusion, where modalities are pre-aligned in a shared latent space. Alexander et al. [36] demonstrated this by using the CLIP architecture to fuse text and images, achieving a high F1-score (97.9%) by leveraging its contrastively

pre-trained shared space. However, recent work by Wan et al. [80] shows that while effective, simpler early fusion models can struggle to process complex syntactic structures, highlighting a performance trade-off.

2. Intermediate Fusion (Cross-Modal Attention): To overcome alignment challenges, the current state-of-the-art has shifted to intermediate (or joint) fusion, where modalities are encoded independently and then fused via cross-attention. Foundational work by Li et al. [81] used this mechanism to fuse text with behavioral metadata, treating statistics as queries to attend to textual values. The potential advantages of this timing were highlighted by Teng et al. [82], whose ablation studies revealed that text-guided cross-attention significantly outperformed direct late concatenation on the clinical AVEC2019 dataset. Further refining this, Shi et al. [37] used audio features as the query and text as the key/value within a transformer decoder, explicitly modeling how vocal hesitation alters textual semantics.

The most effective intermediate fusion strategies, as shown by Jia et al. [83], are preceded by InfoNCE contrastive alignment, which globally aligns the modalities before fine-grained attention is applied. Bridging the gap between linguistic content and temporal patterns, Bin Saeed and Cha [45] proposed DABLNet, which treats temporal metadata (posting hour and day) as a distinct modality. By using a cross-modal attention mechanism to dynamically weight the importance of “what was said” (BiLSTM text features) versus “when it was said” (LSTM temporal features), they achieved a significant performance gain over text-only baselines, demonstrating that behavioral rhythms provide an orthogonal signal essential for early risk detection.

3. Late Fusion (Decision-Level): Late fusion processes modalities through entirely separate deep networks, aggregating only their final

probabilistic outputs. While less common in recent literature due to its inability to capture low-level synergies, this strategy offers superior fault tolerance. De Almeida et al. [84] employed a stacking ensemble where the decisions of independent audio, video, and text models were fused by a meta-learner (BiLSTM), providing a robust approach for handling missing modalities [85].

4. Dynamic and Adaptive Fusion (The 2026 Frontier): The most recent architectural breakthrough is the transition from static to dynamic fusion. Recognizing that modality reliability fluctuates per patient, models now dynamically “gate” or weight modalities in real-time. Anas and Ahmed [86] introduced MAGNet, an emotion-conditioned gating mechanism that acts as a “soft switch” to prioritize text or speech streams. Similarly, Huo et al. [87] and Li et al. [16] proposed architectures that use gating networks to route data through either a lightweight single-modality path or a deep multimodal path depending on the input’s complexity, bridging the gap between rigorous fusion and computational efficiency.

Expanding the sensory scope, foundational work by Xu and Sun [88] integrated deep acoustic features from wav2vec 2.0, confirming that vocal biomarkers provide a critical orthogonal signal that these advanced fusion architectures can leverage. Furthermore, while current multimodal fusion heavily relies on text and images, recent meta-analyses of wearable device data [89] demonstrate high sensitivity (0.89) in detecting physiological depressive markers (e.g., sleep and heart rate variability). Future data-level fusion architectures must bridge the gap between passive linguistic phenotyping (text) and these physiological signals to reduce false positives.

These advanced fusion strategies are now enabling deeper levels of clinical insight. In the domain of parameter-level multimodal fusion, Tahir et al. [90] developed ContextVecNet, utilizing “latent prompt coupling” to project textual prompts into visual space without full fine-tuning. Addressing the interpretability of multimodal models, Zhou et al. [91] utilized a dual-stream Q-former bridge to translate audio and visual cues into an LLM’s prompt space, generating human-readable “Emotion-Cognition Profiles.” Furthermore, multimodal fusion has proven critical for demographic nuance; Jin et al. [92] utilized tri-modal temporal emotion overlap to prove that clinical depression manifests distinctly across genders (e.g., higher acoustic/visual anger in males versus profound sadness in females). Finally, these fusion strategies are also applied at the user level. Rahman et al. [93] used a multi-instance learning (MIL) framework, treating a user as a “bag” of multimodal instances to visualize emotional trajectories over time, effectively acting as a form of interpretability fusion. These diverse technical approaches are summarized in Table 8.

4.4.1. Cross-lingual & low-resource adaptation

The majority of depression detection research is English-centric, creating a “Linguistic Gap” for the global population. Recent work focuses on adaptation fusion, a paradigm that transfers knowledge from high-resource to low-resource languages through code-mixing, translation, and ensemble strategies, as categorized in Table 9.

In many non-Western contexts, users mix languages (e.g., English and Bangla) or scripts. Rahman et al. [93] addressed this by evaluating Multilingual BERT (mBERT) on code-mixed Bangla-English data. They introduced a semantic preprocessing step that fuses emoji translation (converting icons to text) with linguistic features, demonstrating that mBERT (90% accuracy) outperforms language-specific monolingual models in code-mixed environments due to its broader vocabulary alignment. For highly complex, code-mixed languages, fusion remains paramount. Shwetha and Pushpalatha [95] successfully fused embeddings from MuRIL and MentalBERT via cross-attention to classify Major Depressive Disorder in code-mixed Kannada-English text. Similarly, Chitrakala and Senthamizhselvi [96] demonstrated that applying heuristic Gannet Optimization to a FastText-CNN-BiLSTM pipeline achieved 98.6% accuracy on noisy Tanglish (Tamil-English) data, outperforming standard multilingual transformers.

Addressing script variation, Ahmad et al. [97] tackled the complexity of Urdu, which appears in both Roman and Nastaliq scripts. They proposed a combination-based fusion strategy, merging datasets of different scripts and employing a custom attention mechanism with mBERT. This strategy effectively unifies the feature space of disparate scripts, achieving 96% accuracy and proving that script unification is a prerequisite for robust low-resource detection.

For languages with scarce labeled data, Rajderkar and Bhat [78] employed a translation-based augmentation strategy. By fusing original English datasets with machine-translated versions across eight Indian languages (e.g., Hindi, Tamil, Marathi), they created synthetic training corpora. Using an ensemble of LSTM and GRU models on this fused data, they achieved consistent performance ($F1 > 0.90$) across all languages, validating that diagnostic linguistic patterns can be transferred across language families via translation fusion.

Baalawi et al. [51] addressed the morphological complexity of Arabic. They proposed a hybrid ensemble that fuses AraBERT (a domain-adapted transformer) with RNNs (LSTM/GRU). This architecture captures both the rich semantic context of AraBERT and the sequential dependencies of Arabic morphology, achieving 93.03% accuracy. Addressing the “Out-of-Vocabulary” (OOV) problem common in informal social media (slang), Mahendra and Setiawan [8] integrated FastText Feature Expansion into a CNN-GRU pipeline for Indonesian text. By fusing standard embeddings with FastText’s character-level n-grams, they successfully captured semantic meaning from slang and misspelled words, a critical adaptation for low-resource languages lacking robust pre-trained transformers.

While Kurniadi et al. [98] focused on English Reddit data, their comparative analysis of BERT vs. RoBERTa highlights a critical finding for cross-lingual research: RoBERTa’s robust optimization provides better stability and loss convergence. This suggests that for low-resource adaptation, starting with robustly optimized baselines (like XLM-R) is preferable to standard BERT variants.

Beyond linguistic barriers, domain adaptation is crucial for cross-platform generalization. As synthesized by Bhatt et al. [7], models trained on long-form introspective text (e.g., Reddit) often fail catastrophically on short-form platforms (e.g., Twitter or Discord) due to severe domain shift. This necessitates cross-platform data-level fusion to maintain diagnostic accuracy when deploying models in the wild.

4.4.2. Temporal & longitudinal modeling

Depression is not a static state but a dynamic process. Temporal fusion involves integrating the time dimension into the feature space to capture symptom progression and causality.

Lv et al. [29] introduced the concept of life event mining. By extracting specific triggers (e.g., “Job Loss,” “Breakup”) and fusing these discrete events with dense text embeddings, they significantly reduced Early Risk Detection Error (ERDE). This approach moves beyond analyzing what a user feels to why they feel it, integrating causal context into the prediction.

Alhamed et al. [24] focused on the “Diagnosis Point” as a critical temporal anchor. By classifying users based on language usage before and after a self-reported diagnosis, they validated that linguistic markers shift longitudinally (e.g., increased self-focus post-diagnosis). This implies that models must be “Time-Aware,” treating pre- and post-diagnosis posts as distinct semantic distributions. Shen and Paik [42] advanced temporal modeling by fusing RoBERTa embeddings with a novel “Personalized Time-Difference” feature (time elapsed since diagnosis). Unlike absolute timestamps, this relative temporal feature anchors the model to the user’s specific pathology timeline. Their LSTM-based model achieved >99% accuracy, demonstrating that fusing relative temporal distance with semantic content captures the progression of the disorder more effectively than content alone.

The fusion of time-series transformers with semantic embeddings marks the latest temporal frontier. Saini and Sen [94] achieved early detection of anorexia nervosa by fusing a PatchTST time-series model

Table 8

Comparative analysis of multimodal and temporal fusion strategies, highlighting fusion timing.

Study	Modalities	Fusion timing	Key mechanism/Architecture	Best Perf.
Li et al. [16]	Audio + Video	Dynamic	DynMM Gating: Routes data based on input complexity (Mamba+Transformer).	F1: 83.8%
Anas and Ahmed [86]	Text + Speech	Dynamic	Emotion-Aware Gating: Soft-switch prioritization of modalities.	F1: 85.1%
Jia et al. [83]	Text + Audio	Intermediate	Contrastive Global Alignment + Cross-Attention.	F1: 87.4%
Zhou et al. [91]	Text+Audio+Vis	Intermediate	Dual-stream Q-former bridge; Generates interpretative text profiles.	F1: 70.3%
Bin Saeed and Cha [45]	Text + Time	Intermediate	Cross-Modal Attention fusing BiLSTM + Meta-LSTM.	F1: 73.8%
Tahir et al. [90]	Text+Image+ Time	Latent (Early)	ContextVecNet: Latent prompt coupling to fuse CLIP and text.	F1: 96.2%
Alexander et al. [36]	Text + Image	Latent (Early)	CLIP Shared Space + Time2Vec.	F1: 97.9%
Jin et al. [92]	Text+Audio+Vis	Feature-Level	Temporal emotion overlap to capture gender-specific nuances.	Acc: 75.0%
De Almeida et al. [84]	Text+Audio+Vis	Late (Decision)	BiLSTM Meta-Learner stacking independent DNN outputs.	F1: 78.7%
Xu and Sun [88]	Text + Audio	Feature-Level	wav2vec 2.0 + Transfer Learning (TL).	Acc: 93.8%
Saini and Sen [94]	Text + Time Series	Parallel Fusion	PatchTST for temporal trends fused with Longformer for semantics.	F1: 85.2%
Rahman et al. [93]	User + Tweets	Multi-Instance	LSTM Autoencoder for trajectory visualization.	Acc: 95.0%

Table 9

Taxonomy of low-resource and cross-lingual adaptation strategies.

Language	Challenge	Strategy	Key reference	Result
Bangla (Mixed)	Script mixing, Emojis	Preprocessing + mBERT	Rahman et al. [93]	Acc: 90%
Kannada (Mixed)	Code-mixing/Low-Resource	Cross-Attn (MuRIL + MentalBERT)	Shwetha and Pushpalatha [95]	F1: 87.7%
Tamil (Mixed)	Informal text/Code-mixing	FastText + Opt. CNN-BiLSTM	Chitrakala and Senthamizhselvi [96]	Acc: 98.6%
Urdu (Mixed)	Roman/Nastaliq scripts	Script Unification	Ahmad et al. [97]	Acc: 96%
Arabic	Morphological complexity	Ensemble (AraBERT+RNN)	Baalawi et al. [51]	Acc: 93.03%
Indonesian	OOV/Slang handling	FastText Expansion	Mahendra and Setiawan [8]	Acc: 83%
Indian (8)	Data Scarcity	Translation Augmentation	Rajderkar and Bhat [78]	F1 > 90%
Russian	Domain Transfer	PEFT/LoRA Fine-tuning	Kuzmin et al. [73]	F1: 88%
Chinese	Cultural Nuance	Context-Enhanced Fusion	Yan et al. [2]	Acc: 94%

with Longformer semantic outputs, proving that mapping user trajectories is clinically superior to treating posts in isolation. To filter transient noise from longitudinal timelines, Kang et al. [99] introduced ALERT, an adversarial stability-aware routing transformer that explicitly distinguishes persistent pathological depression from temporary emotional distress.

Finally, bridging the gap between computational modeling and clinical epidemiology, Nagata et al. [100] utilized Random-Intercept Cross-Lagged Panel Models (RI-CLPM) on a massive longitudinal cohort ($N=11,876$). Their statistical fusion of time-lagged data proved a unidirectional association: social media use predicts subsequent depressive

symptoms. This provides strong epidemiological support for the temporal fusion efforts in NLP, confirming that historical usage patterns are indeed causal predictors of future mental states.

5. Discussion

This systematic review highlights a fundamental paradigm shift in social media depression detection: the transition from end-to-end deep learning to information fusion. Our analysis of the 2024–2026 literature reveals that the field is no longer defined by isolated feature extraction, but rather by the sophisticated synthesis of neural architectures, symbolic knowledge, and generative reasoning. Although these fusion strategies offer unprecedented semantic capabilities, they also expose a critical divergence between computational metrics and true clinical utility. This section evaluates these trade-offs, contrasting the “illusion of accuracy” prevalent in current benchmarks against the strict requirements for reliable, explainable clinical screening.

5.1. Critical benchmarking: The “Illusion of Accuracy” in benchmarking and the SOTA divide

Importantly, because the surveyed studies utilize disparate datasets, varied feature extraction techniques, and different evaluation protocols, direct head-to-head statistical comparisons are not strictly feasible. The following benchmarking aims to highlight broad architectural trends and reported upper bounds within specific data tiers, rather than establish controlled experimental equivalencies. Furthermore, the aggregation of distinct metrics (e.g., F1-score vs. Accuracy vs. ERDE) across fundamentally different class-imbalance scenarios renders direct comparative ranking mathematically invalid.

Transitioning from qualitative architectural analysis to quantitative evidence, we perform a critical benchmarking of the fusion strategies discussed in Section 4. Table 10 presents the “Grand Benchmark,” consolidating results from the most significant studies of the 2024–2026 period.

5.1.1. The “Illusion of Accuracy” in benchmarking

A critical analysis of the literature reveals a stark divergence in performance metrics driven by data provenance rather than architectural superiority. As detailed in Table 10, studies utilizing Tier 3 proxy-labeled datasets (e.g., Sentiment140, Kaggle-sourced) frequently report accuracies exceeding 99% [e.g., 38,42]. We caution that these hyper-inflated figures are likely statistical artifacts of the data construction process. Specifically, models trained on these datasets often learn to detect negative sentiment polarity (a trivial task) rather than the complex pathology of clinical depression. This phenomenon remains pervasive in 2026; for instance, Balraj and Nagaraj [101] reported an accuracy of 99.68% using an optimized CNN-BiLSTM. However, because the evaluation utilized a small Kaggle dataset lacking true clinical ground truth, this result highlights the persistent risk of complex models overfitting to proxy variables. Similarly, Dhumane et al. [44] achieved an F1-score of 92.9% using their H-Dense FHNNet architecture on a Kaggle-sourced subreddit proxy dataset. This further suggests that highly complex models can overfit to the distinct vocabulary of specific sub-communities rather than learning generalizable clinical patterns.

In contrast, studies utilizing high-validity clinical and expert-annotated datasets (Tier 1 and Tier 2, e.g., DAIC-WOZ, CLEF eRisk) rarely report F1-scores exceeding 0.80 or accuracies above 90%. This lower baseline reflects the genuine difficulty of the clinical screening task and the “ecological validity” of the data. While emerging generative fusion strategies utilizing clinical RAG [35] are beginning to push this clinical ceiling toward 0.95, we argue that “state-of-the-art” remains a dataset-dependent designation. To ensure scientific rigor, comparisons must strictly be made within the same validity tier to avoid conflating clinical screening with trivial sentiment detection.

5.1.2. Reported upper bounds on user-level tasks: The generative vs. Symbolic divide

For user-level analysis, where the goal is to aggregate sparse signals from a timeline into a cohesive diagnosis, two distinct fusion paradigms dominate the literature:

- **Superior Encoding on High-Volume Datasets (LLM Embeddings):** On massive, noisy Twitter datasets, the shift is not from discriminative to generative tasks, but rather from older encoders (BERT) to superior semantic representations derived from modern LLMs. Nareshkumar and Nimala [102] established a robust baseline for architecture fusion, achieving 92% accuracy using a distilled Enhanced BERT coupled with BiLSTM. However, recent architectures have significantly raised the reported upper bounds by using modern LLMs as embedding engines. Shen and Paik [42] reported an exceptionally high accuracy of 99.44% on the Shen et al. dataset using Llama 2 embeddings with a novel “personalized time-difference” feature. This 7% performance jump indicates that for short, context-poor tweets, the massive pre-trained world knowledge of an LLM acts as a powerful semantic feature extractor. It fills in context that standard BERT models miss, even if the final classification is ultimately handled by a lightweight downstream model.
- **Early Risk Detection (Neuro-Symbolic Efficacy):** Conversely, on the rigorous CLEF eRisk benchmark (Reddit), which requires detecting risk early in a user’s timeline, knowledge fusion is highly prevalent. Dou and Kang [11] achieved a state-of-the-art ERDE score of 0.035 (lower is better) using TAM-SenticNet. By fusing a time-aware LSTM with the SenticNet symbolic knowledge base, their model reported favorable metrics compared to their pure neural baselines. This indicates that detecting the onset of depression requires the precision of symbolic logic to identify risk factors (e.g., “insomnia,” “suicide”) that purely statistical models might treat as noise. Chen et al. [62] reinforce this, achieving an F1 score of 0.742 by using a “Three-Way Decision” gate to inject commonsense knowledge (CoKE) only when the neural model is uncertain.

5.1.3. Reported upper bounds on binary classification: The data scale factor

Our analysis reveals that performance on binary classification is not determined solely by architecture, but also by the scale and validity of the training data. We categorize the SOTA results into three distinct tiers based on data volume.

- **Massive Scale (~232k+ posts) and the Proxy Trap:** On massive datasets like “Suicide and Depression Detection,” architecture fusion via massive ensembles frequently yields the highest raw metric scores. For example, Baydili et al. [38] reported a near-perfect accuracy of 99.96% using an ensemble of 8 pre-trained language models (PLMs). However, rather than treating this as a state-of-the-art milestone, our analysis concludes this is a prime example of the “illusion of accuracy”. Achieving near 100% on a Tier 3 dataset indicates that the model has successfully learned to map proxy variables (e.g., negative sentiment polarity) rather than identifying true pathological depression. Therefore, while architecturally complex, such benchmark performance is strictly disconnected from actual clinical deployment readiness.
- **Medium Scale (~7k posts):** On the cleaner “Depression Reddit” dataset, Salameh et al. [103] achieved 96.00% accuracy using DLAD, a hybrid RoBERTa + CNN architecture. This validates the “hybrid” hypothesis: the RoBERTa layer captures global semantic context, while the CNN head is necessary to extract local n-gram patterns (e.g., specific depressive phrases) that dense embeddings might smooth over.

Table 10
Representative performance landscape (2024–2026): Highlighting reported metrics across disparate datasets and fusion strategies.

Dataset (Tier)	Task	Model/Strategy	Metric	Reliability/Validity Note	Reference
Kaggle Survey (Tier 3)	Post Binary	CNN-BiLSTM	Acc: 99.68%	Proxy Risk: Extreme overfit on unverified survey proxy data.	Balraj and Nagaraj [101]
Shen et al. (Tier 2)	Post Binary	LSTM + Llama 2 + Time-Diff	Acc: 99.44%	Self-Report Risk: Despite Tier 2 provenance, highly inflated accuracy suggests model behaves as if trained on Tier 3 proxy data, capturing anchor-tweet artifacts rather than general pathology.	Shen and Paik [42]
Suicide & Dep. (Tier 3)	Post Binary	Ensemble of 8 PLMs	Acc: 99.96%	Artifact: Near-perfect score indicates non-clinical task.	Baydili et al. [38]
CLEF eRisk (Tier 2)	Early Risk	TAM-SenticNet	ERDE: 0.035	High-Quality Benchmark: Validated on user history.	Dou and Kang [11]
	Early Risk	CoKE (Knowledge)	F1: 0.742	Reliable: Neuro-symbolic gating improves robustness.	Chen et al. [62]
Hao Daifu (Tier 1)	Suicide Risk	RAG + Qwen3	F1: 0.954	Gold Standard: Validated by 3 board-certified psychiatrists.	Xie et al. [35]
Reddit Cleaned (Tier 2)	Post Binary	DLAD (RoBERTa+CNN)	Acc: 96.0%	Caution: High post accuracy ≠ user diagnosis.	Salameh et al. [103]
LT-EDI (Tier 2)	Severity	SVM + SMOTE	Acc: 80.0%	Robust: Simpler model beats DL on small expert-annotated data.	Subramanian et al. [17]
E-DAIC (Tier 1)	Binary	BERT + T5 (Parallel)	F1: 0.900	Reliable: High validity clinical interviews.	Wang et al. [46]
WU3D (Tier 2)	User-Level	UMID+ DKE+ MUDD	Acc: 97.45%	SOTA (Chinese): High user-level accuracy via symptom mapping.	Su et al. [39]

Note: Direct statistical comparison between rows is invalid due to heterogeneous data provenance.

- **Small Scale (~1.8k posts):** On scarce datasets (Pirina & Çöltekin), complex transformers often overfit. Tran et al. [104] achieved an impressive F1 score of 0.95 using CLSDepDet, which fuses an LSTM with handcrafted features (HCAN). This highlights a critical finding: when data is scarce, fusing “deep” features with “handcrafted” linguistic priors is superior to relying on end-to-end deep learning alone.

5.1.4. Reported upper bounds on severity estimation: Granularity and imbalance

Moving beyond binary detection to the clinical task of severity estimation (Mild/Moderate/Severe), the primary challenges are class imbalance and the semantic overlap between adjacent severity levels.

- **Fine-Tuning Strategies:** Qasim et al. [28] demonstrated that fine-tuned RoBERTa achieves an F1 score of 0.91 on the Reddit Severity dataset (24k posts). Their ablation study showed that dynamic contextual embeddings (RoBERTa) are strictly necessary to distinguish “Moderate” from “Severe” depression due to the subtle semantic gradation between these classes.
- **The Complexity Trap (Ensembles vs. Classical ML):** The LT-EDI dataset provides a critical case study on the limits of architecture fusion. Le-Hinh et al. [105] proposed a sophisticated hierarchical ensemble of five different transformers (BERT, RoBERTa, XLNet), yet achieved only 66.8% accuracy due to the dataset’s small size (~7000 samples). In stark contrast, Subramanian et al. [17] reported that a classical SVM with SMOTE achieved 80% accuracy

on the same data. This comparison serves as a clear indication: complex information fusion strategies induce overfitting on small datasets, where simpler discriminative boundaries (SVM) are more robust.

- **Fusion with Augmentation:** Addressing this data scarcity, Kerasiotis et al. [27] utilized the Depression Severity Dataset and achieved a weighted F1 score of 84.15%. Their strategy fused DistilBERT embeddings with metadata (medication mentions) and easy data augmentation (EDA). This suggests that for fine-grained tasks, data augmentation is often a highly effective alternative to complex architectural fusion.

5.2. Open challenges & strategic future directions

Despite the rapid architectural advancements detailed in Section 4.1, the field of social media depression detection faces a “deployment gap.” While offline metrics (F1-scores, Accuracy) have reached near-perfection on specific benchmarks, the translation of these models into clinical practice remains stalled by fundamental issues regarding data validity, computational feasibility, and ethical safety. This section outlines the four critical barriers identified in the 2024–2026 literature and proposes strategic directions for future information fusion research.

5.2.1. The “Illusion of Accuracy” & the data validity crisis

A recurring pattern in recent literature is the reporting of near-perfect accuracies (> 99%) on datasets constructed via proxy labeling.

- **The Challenge:** As highlighted in our analysis of Abuhasirah [106], studies utilizing the Sentiment140 dataset frequently

achieve accuracies exceeding 99.9% by treating “negative sentiment” as synonymous with “depression”. This represents a methodological fallacy. Phiri et al. [4], in their meta-analysis, confirm that effect sizes are artificially inflated in studies lacking clinical ground truth. The model learns to detect sadness (an emotion) rather than depression (a pathology). Furthermore, regarding data governance, Appendix (Table A.1) highlights that while studies utilizing shared tasks (e.g., CLEF eRisk) legitimately operate under “Shared-Task / Exempt” status, numerous studies involving primary data scraping fail to explicitly report IRB approval.

- **Fusion Implication:** In an information fusion context, fusing a high-variance noisy source (like proxy-labeled tweets) with a robust source (like clinical questionnaires) often degrades the performance of the robust source.
- **Strategic Direction (Clinical-Grade Fusion):** Future research must move toward weak supervision fusion. Rather than treating all training data as equal, models should employ “data programming,” where clinically annotated datasets (Tier 1) serve as the labeling functions to re-weight or filter noisy, large-scale social data (Tier 3). Mendes and Caseli [66] demonstrate the viability of this by focusing on fine-grained symptom detection (e.g., sleep disorders) rather than binary labels, a strategy that aligns computational outputs with diagnostic criteria.

5.2.2. The cost of fusion vs. Real-time deployment

There is a diverging trend between “Research SOTA” and “Deployable SOTA”.

- **The Challenge:** The ensemble proposed by Noor et al. [10] (SENS-Deep) stacks six independent transformer architectures, and Baydili et al. [38] utilize an ensemble of eight pre-trained language models. While these approaches achieve statistical supremacy, their inference latency and memory footprint render them undeployable in real-time monitoring scenarios or on edge devices (e.g., smartphones).
- **The Trade-off:** Ding et al. [58] quantified this trade-off, revealing that while deep ensembles offer marginal F1 gains (+0.02), they incur a computational cost over 3000× higher than optimized classical baselines.
- **Strategic Direction (Distillation & Pruning):** The future lies in knowledge distillation fusion, where heavy ensembles (teachers) are distilled into lightweight student models suitable for edge deployment. Nongkynrih et al. [60] provided a proof-of-concept by deploying an optimized SVM via a mobile Flask API, proving that architectural simplicity is a prerequisite for accessibility. Future work should focus on distilling the reasoning capabilities of LLMs into compact networks (e.g., DistilBERT or TinyLlama) to balance the reasoning-latency trade-off.

5.2.3. Privacy, data governance, and the API dilemma

The shift toward generative fusion (Section 4.3) introduces severe privacy and compliance risks that were less prevalent in the era of static embeddings.

- **The Challenge (The API Data Residency Risk):** A critical vulnerability in current generative fusion research is the reliance on commercial “black-box” APIs (e.g., OpenAI’s GPT-4), as utilized by Shah et al. [40] and Lan et al. [13]. Unlike local model deployments, API-based architectures necessitate transmitting sensitive user text to third-party servers for inference. For clinical applications, this raises significant data residency and GDPR compliance issues, as health data effectively leaves the secure clinical environment. Owen et al. [5] highlight that standard user consent models are currently insufficient for this new paradigm of third-party generative profiling.

- **Technical Mitigation:** Recent work attempts to obfuscate data before transmission. Eswar et al. [107] introduced a novel privacy-preserving fusion layer using discrete Fourier transforms (DFT). By converting embeddings into the frequency domain and injecting Gaussian noise, they obfuscate the semantic readability of the text while retaining the structural patterns necessary for classification.
- **Strategic Direction (Local & Federated Paradigms):** To bridge the deployment gap, future architectures must shift away from API dependency toward distilled local models. Research should focus on distilling the reasoning capabilities of massive LLMs into compact, edge-deployable networks (e.g., TinyLlama or DistilBERT) that process data entirely within the user’s device or clinical server. Furthermore, adopting federated learning, where fusion occurs at the edge and only gradient updates are shared, offers a compliant path forward. Combining these local architectures with differential privacy layers will be essential to prevent the reconstruction of user identities from model outputs.
- **Dataset Provenance and Ethical Compliance:** As detailed in Appendix (Table A.1), we categorized the ethical status of the surveyed studies into standardized labels. Studies utilizing established shared-task benchmarks (e.g., CLEF eRisk, LT-EDI) are classified as “Shared-Task / Exempt,” as primary data collection and anonymization were handled by task organizers. Similarly, studies utilizing established, de-identified clinical benchmarks (e.g., DAIC-WOZ) are categorized as “Secondary Analysis Exempt”. However, for studies involving independent web scraping (primarily Tier 2 and Tier 3 datasets), we verified the presence of explicit ethical review. Our analysis reveals that while Tier 1 clinical studies consistently report formal “IRB Approved” status, studies relying on scraped proxy data frequently fail to report any ethical oversight, relying implicitly on the assumption that public data equates to ethical clearance. This highlights a persistent gap in data governance for large-scale generative fusion.

5.2.4. The hallucination risk & neuro-symbolic trust

While LLMs offer reasoning, they lack reliability.

- **The Challenge:** Zheng et al. [14] observed that chain-of-thought (CoT) prompting degraded performance because the LLM hallucinated symptoms in neutral texts. Similarly, Sathvik et al. [76] found that zero-shot LLMs struggle to distinguish between “demotivation” and clinical “depression,” leading to high false-positive rates.
- **Fusion Implication:** Unconstrained generative fusion is unsafe for clinical screening.
- **Strategic Direction (Graph-Guided Verification):** The optimal path forward is graph-guided generative fusion. As demonstrated by Mao and Han [108] with TextGCN and Zhang et al. [12] with Depression Ontologies, structured knowledge graphs should act as a “fact-checking” layer. Future systems should use LLMs to generate hypotheses (e.g., “User shows signs of insomnia”), which are then verified against a clinical knowledge graph before being accepted by the fusion engine. This neuro-symbolic verification loop ensures that the model’s “reasoning” remains bounded by established medical logic.

To consolidate the critical barriers and forward-looking solutions discussed throughout this section, we present a strategic roadmap in Table 11.

5.3. Pathways to clinical integration and ethical deployment

While Sections 4 and 5.1 detail the rapid technical evolution of information fusion, the 2025–2026 literature reveals a growing consensus that algorithmic superiority alone is insufficient for real-world

Table 11
Strategic roadmap for information fusion in depression detection.

Domain	Current limitation	Evidence	Strategic direction
Data Validity	“Illusion of Accuracy” on proxy labels (Sentiment $\neq$ Depression).	Abuhasirah [106], Phiri et al. [4], Tang et al. [30]	Weak Supervision Fusion: Use clinical data to re-weight noisy social signals [31].
Deployment	Massive ensembles are computationally undeployable.	Noor et al. [10], Ding et al. [58]	Distillation Fusion: Distilling LLM reasoning into lightweight Edge models.
Privacy	API-based LLMs expose sensitive user data.	Owen et al. [5], Shah et al. [40], Thamrin and Chen [109]	Obfuscated Fusion: Fourier Transforms [107] and Differential Privacy [110].
Clinical Use	AI viewed as a replacement tool, causing “technological solutionism” friction.	Ghorbankhani and Safara [20]	Complementary Triage: Deploy as conversational agents supporting clinicians.
Ethics	Massive lack of IRB approval and informed consent in dataset creation.	Bucur et al. [34], Wright et al. [19]	Governance: Mandate TRIPOD+AI reporting guidelines.
Reliability	LLMs hallucinate symptoms in neutral text.	Zheng et al. [14], Sathvik et al. [76], Yang et al. [21]	Neuro-Symbolic Verification: Using Knowledge Graphs [108] and RAG [35] to audit outputs.
Structure	Text-only models miss network effects.	Mao and Han [108]	Graph-Text Fusion: Integrating TextGCN to capture community homophily.

medical deployment. As highlighted by comprehensive clinical reviews [19,20], the field must address practical considerations regarding how these models complement existing healthcare infrastructure, protect user privacy, and navigate complex ethical landscapes.

5.3.1. Complementarity with existing clinical screening tools

A critical barrier to adoption is the risk of “technological solutionism,” which is the misconception that AI should replace traditional psychiatric assessments (e.g., PHQ-9, BDI-II). Instead, recent literature advocates for deploying fusion models as clinical decision support systems (CDSS) that complement rather than replace human judgment [20]. For example, García-Méndez et al. [111] demonstrated a highly practical implementation via the “Celia” chatbot. Rather than outputting a sterile binary diagnosis, this conversational assistant longitudinally monitors elderly users, using an LLM to generate natural language explanations of potential anxiety or depression markers, which are then passed to human clinicians for review. Similarly, Wan et al. [80] proposed integrating RoBERTa-GRU fusion networks directly into university digital counseling portals to trigger early, proactive interventions. In these complementary frameworks, the AI acts as a high-throughput, continuous triage system, flagging nuanced distress signals that might otherwise remain hidden between episodic clinical visits. However, clinical integration faces hurdles regarding diagnostic specificity. For instance, current wearable AI models exhibit low specificity (0.65) for predicting future depressive episodes due to inherent physiological data noise [89]. To achieve the robust specificity required for clinical validation, future screening tools must fuse the contextual neuro-symbolic reasoning derived from social media text with the objective physiological ground truth of wearable sensors [79].

5.3.2. Privacy-preserving AI and the API dilemma

The shift toward generative fusion introduces severe privacy and HIPAA/ GDPR compliance risks. Unlike local models, API-based architectures (e.g., OpenAI’s GPT-4) necessitate transmitting sensitive user text to third-party servers. To bridge this deployment gap, researchers are actively developing privacy-preserving fusion strategies. Beyond early proposals like Fourier transform obfuscation [107], concrete implementations are now emerging. A prime example is the DP-CARE framework by Karpontinis and Soufleri [110], which fuses a frozen transformer encoder with a lightweight classifier trained under strict differential privacy (DP) guarantees. Their results prove that a robust

privacy budget ($\epsilon \approx 3$) can be maintained with only a minimal drop in F1-score (from 80.5% to 78.1%), while critically improving recall for stress detection. Furthermore, Thamrin and Chen [109] demonstrated that utilizing fine-tuned, localized open-weight LLMs (e.g., MentalLlama) entirely avoids the data-leakage risks associated with commercial APIs while simultaneously reducing clinical hallucinations.

5.3.3. Ethical review and the scarcity of data governance

Perhaps the most significant practical challenge is the persistent lack of robust ethical governance in training data collection. A massive 2026 survey of 108 non-English mental health datasets by Bucur et al. [34] revealed that only approximately 16% of recent studies reported formal Institutional Review Board (IRB) approval. The vast majority relied on a “public domain” justification for scraping deeply sensitive mental health disclosures without explicit user consent. Furthermore, models trained on such unverified data often conflate transient negative sentiment with clinical stigma or actual pathology [19]. For clinical implementation to become a reality, rigid adherence to ethical data governance must transition from an exception to the standard. Future fusion architectures must be evaluated not merely on F1-scores, but on their adherence to established clinical reporting standards like TRIPOD+AI [23], ensuring transparency, accountability, and the ethical safeguarding of vulnerable populations.

5.4. Limitations of the review

While this systematic review offers a rigorous synthesis of the pivotal 2024–2026 period, the scope and conclusions are subject to several inherent limitations. First, our search strategy was intentionally focused on six primary repositories spanning computer science and interdisciplinary medicine (IEEE Xplore, ACM DL, Scopus, SpringerLink, PubMed/Medline, and APA PsycINFO). By design, while this covers top-tier venues, it may underrepresent highly specialized clinical niche journals. Second, the search query was conducted in English, acting as a linguistic filter that may exclude relevant, high-impact research published natively in other languages. Although we included and analyzed multilingual studies, their initial discovery was contingent on English-language metadata.

Crucially, the findings of this review are constrained by the methodological heterogeneity and data provenance of the primary studies themselves. As our analysis of the “Illusion of Accuracy” highlights

(Section 5.1.1), the performance metrics reported in the literature are heavily influenced by dataset validity. Therefore, our synthesis reflects the state of the published field, including the prevalent issue of inflated performance on low-validity, proxy-labeled datasets. Additionally, as with all such reviews, this work is susceptible to publication bias, wherein studies reporting novel models with statistically significant, positive results are more likely to be published than those with null or negative findings, potentially leading to an overestimation of the general efficacy of the fusion strategies discussed.

Finally, as noted in recent comprehensive meta-analyses [30,31], our cross-study comparisons (e.g., the performance heatmap) aggregate heterogeneous tasks and metrics; therefore, they represent broad field-wide trends rather than controlled experimental equivalencies.

6. Conclusion

This survey systematically analyzes the rapid evolution of social media depression detection during the pivotal years of 2024–2026. Our review of 87 key studies reveals a fundamental paradigm shift: the field has graduated from the era of static end-to-end deep learning into the era of information fusion. By moving beyond simple binary classification, recent research has redefined the task from identifying what a user is feeling to understanding why they feel it, marking a transition from detecting surface-level sentiment to modeling true clinical pathology.

Our critical benchmarking highlights a distinct dichotomy in the current state-of-the-art. Architecture fusion, exemplified by massive heterogeneous ensembles [10,38], frequently reports exceptionally high accuracies (> 99%) on general benchmarks. However, our synthesis indicates that this performance often relies on proxy-labeled data and incurs computational costs that render deployment on edge devices unfeasible. In contrast, generative fusion offers unprecedented semantic capabilities for interpreting short, sparse text [42]. Yet, recent empirical benchmarks suggest that without task-specific parameter adaptation (PEFT), zero-shot large language models perform poorly on nuanced clinical tasks and remain vulnerable to severe hallucination risks [59,74].

Consequently, current evidence suggests that a highly promising direction for the field lies in bridging advanced multimodal data-level fusion with knowledge-augmented fusion, specifically through neuro-symbolic architectures. Models that dynamically gate multimodal signals [16] and fuse neural embeddings with structured clinical ontologies [12] or retrieval-augmented generation (RAG) [35] currently appear to offer a highly effective balance of robustness, interpretability, and clinical validity. Looking forward, we recommend a strategic pivot away from ever-larger general-purpose large language models. Instead, future research should focus on developing compact, clinically grounded models. This includes distilling the reasoning capabilities of LLMs into lightweight networks [60], employing formal privacy-preserving mechanisms such as differential privacy [110], and mandating adherence to clinical reporting standards like TRIPOD+AI [23]. Ultimately, the goal of information fusion in this domain is not merely to detect signals in the digital noise, but to provide the explainable, trustworthy insights required to trigger life-saving clinical interventions.

CRedit authorship contribution statement

Zeynab Davoudi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Meysam Roostae:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Khadijeh Aghajani:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Ethics statement

This research is a systematic review of existing published literature and did not involve any experiments with human or animal subjects; therefore, no ethical approval was required.

Declaration of generative AI in scientific writing

The manuscript text has been reviewed using AI-powered proofreading tools. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the Department of Computer Engineering at the University of Mazandaran for providing the necessary research environment and resources to conduct this systematic review. We also express our gratitude to the primary researchers of the pivotal studies analyzed in this review, whose work forms the foundation of our analysis. Furthermore, we extend our sincere appreciation to the handling editor and the anonymous reviewers, whose insightful and constructive feedback significantly elevated the scope, clinical depth, and methodological rigor of this manuscript. No specific funding was received from any agency in the public, commercial, or not-for-profit sectors for this research.

Appendix. Comprehensive list of surveyed studies and quality assessment

Table A.1 presents the complete catalog of the 87 primary studies analyzed in this systematic review. To prevent double-counting across our narrative synthesis, each study in this appendix is assigned a single, mutually exclusive fusion paradigm based on the hierarchical tie-breaker rule established in Section 2.2 (Data-Level → Generative → Knowledge → Architecture). To address the critical need for methodological transparency, each study was evaluated on its computational fusion paradigm, dataset validity tier, ethical compliance (e.g., formal IRB approval or authorized secondary use), and methodological rigor.

Data extraction and quality assessment were performed independently by two reviewers, with any discrepancies resolved through consensus discussion. Informed by the principles of the TRIPOD+AI clinical reporting guidelines [23], each study was categorized on a custom three-point rubric designed to evaluate the synergy between data provenance and experimental design:

- **High Rigor (3/3):** Studies utilizing Tier 1 or high-quality Tier 2 clinical/expert-annotated datasets, featuring comparative base-lines, and providing sufficient ablation or robust evaluation protocols.
- **Medium Rigor (2/3):** Studies utilizing self-reported data with standard architectural evaluations but lacking deeper feature ablation or direct clinical validation.
- **Low Rigor (1/3):** Studies utilizing Tier 3 proxy-labeled data with minimal ablation, flagged for high susceptibility to the “illusion of accuracy”.

This assessment highlights whether reported performance metrics are computationally sound and clinically reliable.

Table A.1
Comprehensive catalog of surveyed information fusion studies (2024–2026).

Reference	Fusion paradigm	Dataset (Tier)	Quality/Rigor	Ethical status
Generative Fusion (LLMs & Prompting)				
AlSaad et al. [74]	Generative (PEFT vs ICL)	Twitter (Tier 2)	High (Robust LLM eval)	Shared-Task/Exempt
Chen et al. [85]	Generative + Multimodal	DAIC-WOZ (Tier 1)	High (Missing mod ablation)	Secondary Analysis Exempt
Xie et al. [35]	Generative (RAG)	Hao Daifu (Tier 1)	High (Psychiatrist scored)	IRB Approved
Lam et al. [70]	Generative (PEFT)	Clinical (Tier 1)	High (Comorbidity analyzed)	IRB Approved
Teng et al. [82]	Generative (Prefix Tune)	AVEC2019 (Tier 1)	High (Cross-Attn ablation)	Secondary Analysis Exempt
Kallstenius et al. [59]	Generative vs Classical	Reddit (Tier 2)	High (Large-scale eval)	Not Reported (Public Data)
Thamrin and Chen [109]	Generative (XAI)	Twitter (Tier 2)	Medium (Hallucination metric)	Not Reported (Public Data)
Zheng et al. [14]	Generative (Cascade)	Weibo (Tier 2)	High (CoT ablation)	Not Reported (Public Data)
Lan et al. [13]	Generative–Discriminative	Reddit (Tier 2)	High (XGBoost validation)	Shared-Task/Exempt
Shah et al. [40]	Generative (LoRA)	Twitter (Tier 3)	Low (Proxy inflation)	Not Reported
Shen and Paik [42]	Generative (Llama2+Temp)	Twitter (Tier 2)	Low (Proxy-like behavior)	Not Reported (Public Data)
Kuzmin et al. [73]	Generative (Russian LLM)	Mixed (Tier 2)	High (Resource trade-off)	Not Reported (Public Data)
Wang et al. [75]	Generative (Zero-Shot)	Reddit (Tier 2)	Medium (BDI evaluation)	Not Reported (Public Data)
Wang et al. [26]	Generative (RAG Timeline)	Reddit (Tier 2)	High (Longitudinal logic)	Not Reported (Public Data)
Sathvik et al. [76]	Generative (Zero-Shot)	M-HELP (Tier 2)	High (Help-seeking intent)	Shared-Task/Exempt
Knowledge & Neuro-Symbolic Fusion				
Wan et al. [69]	Knowledge (CBT injection)	SWDD (Tier 2)	High (Clinical pre-training)	IRB Approved
Zhang et al. [65]	Knowledge (Loss function)	MODMA (Tier 1)	High (Cross-dataset tests)	Not Reported (Public Data)
Lissak et al. [71]	Knowledge (Path Analysis)	Facebook (Tier 1)	High (CSSRS validated)	IRB Approved
Dou and Kang [11]	Knowledge (SenticNet)	eRisk (Tier 2)	High (Early risk focus)	Shared-Task/Exempt
Zhang et al. [12]	Knowledge (Ontology)	Reddit (Tier 2)	High (Dual-attention)	Shared-Task/Exempt
Chen et al. [62]	Knowledge (COMET graph)	eRisk (Tier 2)	High (Uncertainty gating)	Shared-Task/Exempt
Mao and Han [108]	Knowledge (TextGCN Graph)	Twitter (Tier 2)	High (Graph-guided context)	Not Reported (Public Data)
Wang et al. [64]	Knowledge (Scale guided)	Weibo (Tier 2)	High (End-to-End BDI)	Not Reported (Public Data)
Su et al. [39]	Knowledge (Dynamic mem)	WU3D (Tier 2)	High (Memory updates)	Not Reported (Public Data)
Mendes and Caseli [66]	Knowledge (Fine-grained)	Reddit (Tier 2)	Medium (Somatic focus)	Not Reported (Public Data)
Farrique et al. [67]	Knowledge (Rare symptom)	Twitter (Tier 2)	High (Semi-supervised LLM)	Not Reported (Public Data)
Data-Level, Temporal & Multimodal Fusion				
Chen et al. [15]	Multimodal (Text+Audio+Vis)	DAIC-WOZ (Tier 1)	High (Clinical outpatient)	IRB Approved
Shi et al. [37]	Multimodal (Transformer)	EATD (Tier 1)	High (Audio-Query Attn)	IRB Approved
Jin et al. [92]	Multimodal (Temporal)	DAIC-WOZ (Tier 1)	High (Gender analysis)	Not Reported (Public Data)
Jia et al. [83]	Multimodal (Contrastive)	EATD (Tier 1)	High (Global alignment)	Not Reported (Public Data)
Bin Saeed and Cha [45]	Multimodal (Text + Time)	Reddit (Tier 2)	Medium (Attention ablation)	Not Reported (Public Data)
Huo et al. [87]	Multimodal (Adaptive)	DAIC-WOZ (Tier 1)	High (Emotion gating)	Not Reported (Public Data)
Anas and Ahmed [86]	Multimodal (Gating)	EATD (Tier 1)	High (Modality switching)	Not Reported (Public Data)
Zhang et al. [77]	Multimodal + Generative	FJSI (Tier 1)	High (Tree-of-Thoughts)	IRB Approved
Li et al. [16]	Multimodal (Dynamic)	D-Vlog (Tier 2)	High (Mamba-Transformer)	Not Reported (Public Data)
De Almeida et al. [84]	Multimodal (Stacking)	EDAIC (Tier 1)	High (CDV generalization)	Not Reported (Public Data)
Zhou et al. [91]	Multimodal (Captioning)	MMDA (Tier 1)	High (Q-former bridge)	IRB Approved
Tahir et al. [90]	Multimodal (Context Vect)	Twitter (Tier 2)	Medium (Latent coupling)	Not Reported (Public Data)
Fu et al. [72]	Data-Level (Transfer)	Breast Cancer (Tier 1)	High (SDS validated)	IRB Approved
Shwetha and Pushpalatha [95]	Data-Level (Cross-Lingual)	Kannada/Eng (Tier 2)	High (Code-mixed focus)	Not Reported
Yan et al. [2]	Data-Level (Cultural Adapt)	Weibo (Tier 2)	High (Emotion-enhanced)	Not Reported (Public Data)
Rajderkar and Bhat [78]	Data (Translation Aug)	Indian Langs (Tier 2)	Medium (Translation bias)	Not Reported (Public Data)
Kerasiotis et al. [27]	Data (EDA Augmentation)	Reddit (Tier 2)	Medium (Severity balance)	Not Reported (Public Data)
Alexander et al. [36]	Multimodal (CLIP Latent)	Twitter (Tier 2)	High (Latent alignment)	Not Reported (Public Data)
Li et al. [81]	Multimodal (Cross-Attn)	Weibo (Tier 2)	Medium (Metadata queries)	Not Reported (Public Data)
Xu and Sun [88]	Multimodal (wav2vec)	DAIC-WOZ (Tier 1)	High (Acoustic vectors)	Shared-Task/Exempt
Rahman et al. [93]	Data (MIL Trajectory)	Bangla Mixed (Tier 2)	High (Trajectory tracking)	Not Reported (Public Data)
Lv et al. [29]	Data (Life Event Mining)	PsyEvent (Tier 2)	High (Psychological triggers)	Shared-Task/Exempt
Alhamed et al. [24]	Temporal (Longitudinal)	Twitter (Tier 2)	High (Pre/Post diagnosis)	Not Reported (Public Data)
Saini and Sen [94]	Temporal (Time-Series)	eRisk (Tier 2)	High (PatchTST trajectory)	Shared-Task/Exempt
Nagata et al. [100]	Temporal (Epidemiological)	Cohort (Tier 1)	High (RI-CLPM metrics)	IRB Approved
Ahmad et al. [97]	Data (Script Unification)	RUDA (Tier 2)	High (Nastaliq/Roman Urdu)	Not Reported (Public Data)
Architecture Fusion (Hybrids & Ensembles)				
Oulahbib et al. [48]	Arch (Multi-Task)	Reddit (Tier 2)	High (Efficiency scaling)	Not Reported (Public Data)
Sarkar and Majumder [54]	Arch (Transformer-RNN)	Reddit (Tier 2)	Medium (Multi-label)	Shared-Task/Exempt
Balraj and Nagaraj [101]	Arch (CNN-BiLSTM)	Kaggle (Tier 3)	Low (Extreme proxy inflation)	Not Reported
Kasap et al. [56]	Arch (Ensemble)	DepressionEmo (Tier 2)	Medium (Voting stacking)	Not Reported (Public Data)
Hsieh et al. [41]	Arch (Prompt Ensemble)	MultiWD (Tier 2)	High (Self-supervision)	Not Reported (Public Data)
Hossain et al. [63]	Arch (Hybrid + Lexicon)	Kaggle (Tier 3)	Low (Proxy dataset)	Not Reported
Ta et al. [25]	Arch (ML Analysis)	Twitter (Tier 3)	Medium (Standard baseline eval)	Not Reported (Public Data)
Chitrakala and Senthamizhselvi [96]	Arch (Opt. CNN-BiLSTM)	Tanglish (Tier 2)	High (Human annotation)	Not Reported (Public Data)
Arshad et al. [57]	Arch (Soft-Voting)	Multi-Platform (Tier 3)	Medium (Cross-platform)	Not Reported (Public Data)
Wang et al. [46]	Arch (Parallel LLMs)	E-DAIC (Tier 1)	High (Logical AND fusion)	Not Reported (Public Data)

(continued on next page)

Table A.1 (continued).

Reference	Fusion paradigm	Dataset (Tier)	Quality/Rigor	Ethical status
Dhumane et al. [44]	Arch (Harmonic Fusion)	Kaggle (Tier 3)	Low (Proxy inflation)	Not Reported
Karponitis and Soufleri [110]	Arch (Privacy-Preserving)	Dreaddit (Tier 2)	High (Differential Privacy)	Not Reported (Public Data)
Kang et al. [99]	Arch (Temporal Routing)	eRisk (Tier 2)	High (Adversarial learning)	Shared-Task/Exempt
Wan et al. [80]	Arch (RoBERTa-GRU)	Reddit (Tier 2)	High (Syntax complexity)	Not Reported (Public Data)
García-Méndez et al. [111]	Arch (Streaming ML+LLM)	Celia Chatbot (Tier 1)	High (Longitudinal study)	IRB Approved
Bouktif et al. [47]	Arch (BERT-CNN-LSTM)	Reddit (Tier 2)	Medium (LIME XAI)	Not Reported (Public Data)
Villa-Pérez et al. [50]	Arch (Early vs Neural)	CLPsych (Tier 1)	High (Temporal split)	Shared-Task/Exempt
Noor et al. [10]	Arch (Stacking)	Kaggle (Tier 3)	Low (Prohibitive scale/proxy)	Not Reported (Public Data)
Mahendra and Setiawan [8]	Arch (CNN-GRU+FastText)	Twitter (Tier 2)	Medium (Slang handling)	Not Reported (Public Data)
Thekkekara et al. [9]	Arch (CNN-BiLSTM-Attn)	eRisk (Tier 2)	High (Interpretability)	Shared-Task/Exempt
Tejaswini et al. [43]	Arch (FastText+CNN-LSTM)	Twitter (Tier 3)	Low (Proxy dataset)	Not Reported (Public Data)
Baalawi et al. [51]	Arch (AraBERT-RNN)	Nafsany (Tier 2)	High (Low-resource Arabic)	Not Reported (Public Data)
Louardi et al. [52]	Arch (RoBERTa)	Suicide Watch (Tier 2)	Medium (Baseline comp)	Not Reported (Public Data)
Raj et al. [53]	Arch (Transformers+LSTM)	Twitter (Tier 2)	Medium (Data size ablation)	Not Reported (Public Data)
Tiwari et al. [55]	Arch (E-CLS)	DepDs (Tier 2)	Medium (Sparse + Deep)	Not Reported (Public Data)
Baydili et al. [38]	Arch (PLM Ensemble)	Kaggle (Tier 3)	Low (Proxy inflation)	Not Reported (Public Data)
Abuhasirah [106]	Arch (Classical Data Mining)	Sentiment140 (Tier 3)	Low (Sentiment conflation)	Not Reported (Public Data)
Ding et al. [58]	Arch (Classical vs Deep)	Reddit (Tier 2)	High (Efficiency metrics)	Not Reported (Public Data)
Nongkynrih et al. [60]	Arch (SVM Edge)	Twitter (Tier 3)	Low (Edge deployment)	Not Reported (Public Data)
Hameed et al. [61]	Arch (XAI + RF)	Kaggle (Tier 3)	Medium (LIME integration)	Not Reported (Public Data)
Kurniadi et al. [98]	Arch (BERT vs RoBERTa)	Reddit (Tier 2)	Medium (Optimization stable)	Not Reported (Public Data)
Salameh et al. [103]	Arch (RoBERTa-CNN)	Reddit (Tier 2)	Medium (Medium scale)	Not Reported (Public Data)
Tran et al. [104]	Arch (LSTM+HCAN)	Pirina (Tier 2)	Medium (Handcrafted priors)	Not Reported (Public Data)
Qasim et al. [28]	Arch (Severity Context)	Reddit (Tier 2)	Medium (Context gradation)	Not Reported (Public Data)
Sheng et al. [49]	Arch (Self-Attn Hybrid)	EATD/CMDC (Tier 1)	High (POS rule injection)	Not Reported (Public Data)

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